EmulatorEggs

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1. Welcome to EmulatorEggs!

Wiki Link

1.1 What is EmulatorEggs?

EmulatorEggs is meant to be a catch-all wiki for emulation on Linux with an eye for the Steam Deck and immutable operating systems. This wiki will cover how to install and configure various emulators using universal Linux packaging such as AppImages and Flatpaks. When possible, pages will adhere to the developer's preferred packaging format. When a packaging format does not exist, a page will include a disclaimer and helpful tips when asking for support.

If an emulator only exists on Windows, this wiki will cover instructions on how to install the emulator through Wine/Proton with heavy emphasis that support for these emulators is not guaranteed when asking for help in official channels.

The goals of the wiki are as follows:

- To cover basic emulation information such as expected BIOS and ROM file types.
- To teach a user how to install and configure an emulator for the first time.
- To teach a user how to navigate to an emulator's configuration folders on Linux with instructions geared towards people who have never used a Linux distro before.
- To teach a user how to interact with configuration files
- To teach a user how to interact with a terminal.
- To teach a user how to augment an emulator's features (example: enable gyroscope).
- To extend the documentation available for emulator developers.

1.2 Contributions

EmulatorEggs is always looking for contributors to add new wiki pages and enhance the documentation. This wiki is hosted on a public GitHub page, pull requests welcome.

1.3 Steam Deck and Game Mode

Many of the sections on this wiki will cover "Game Mode" specific steps. These steps will only be applicable if you are using a Steam Deck, a distro with "Game Mode" support (Bazzite, Nobara, Manjaro, etc.), or if you have installed "Game Mode" on your own (through the Gamescope package, https://github.com/ValveSoftware/gamescope).

1.4 Front-Ends

This wiki will cover how to configure and set up front-ends which can be used to launch and manage your ROMs. Sections may include specific steps on how to configure an emulator for a front-end.

If you have created a front-end or would like to see a front-end page, pull requests are welcome to cover additional front-ends on the wiki.

1.5 Disclaimer

This wiki does not seek to be a replacement for official emulation documentation but rather fill in the gaps. When possible, pages will link to official documentation and encourage the use of the original documentation when possible.

Emulator developers are more than welcome to take the work here and use it in their own documentation.

May 27, 2026

November 16, 2025

2. First Steps

2.1 First Steps Table of Contents

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The Basics

$HOME or ~ refer to your home directory. On a typical linux distro, the full path will be $HOME/$USER. On a Steam Deck, the full path will be `/home/deck`. When navigating using the file manager, the file manager may not be display the username (the username being `deck` on a Steam Deck) visually in the full file path.

Directories with a `.` in front of their name, like `$HOME/.steam`, are hidden by default. In Dolphin (a file manager), click the hamburger menu in the top right and toggle on `Show Hidden Files`. The setting may vary depending on which file manager you are using.

$HOME/Applications refers to the Applications folder in your home directory. On a typical linux distro, the full path will be $HOME/$USER/Applications. On a Steam Deck, the full path will be `/home/deck/Applications`. When navigating using the file manager, the file manager may not display the username (the username being `deck` on a Steam Deck) visually in the full file path. The Applications folder is not created by default. If you have not created one already, do so before following the steps in this section or on this page.

2.2 How to Set up a Basic Emulation Directory

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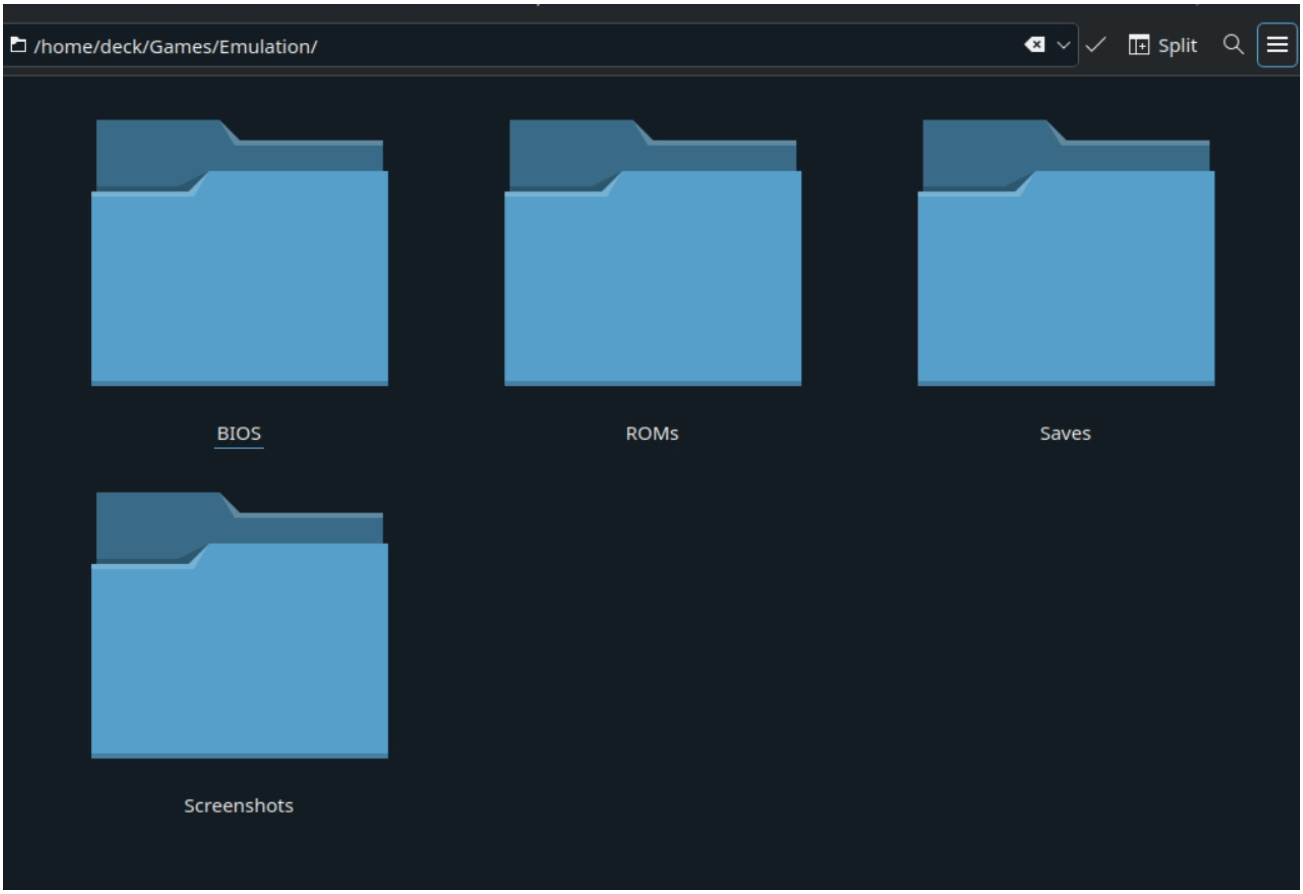
> **Info**
>
> This page assumes you will use the paths in the section below. You are not required to use these paths. However, it is recommended you keep the folder names for consistency across the sections on this page. For example, if you are using an external storage device, it is recommended you use something like `/run/media/deck/Emulation` as your base folder.

1. In the `$HOME` folder, create a `Games` folder if one does not exist ⓘ
   - $HOME or ~ refer to your home directory. On a typical linux distro, the full path will be $HOME/$USER. On a Steam Deck, the full path will be `/home/deck`. When navigating using the file manager, the file manager may not be display the username (the username being `deck` on a Steam Deck) visually in the full file path.
2. In the `$HOME/Games` folder, create an `Emulation` folder
3. In the `$HOME/Games/Emulation` folder, create the following folders:
   - `BIOS`
   - `ROMs`
   - `Saves`
     - In the `Saves` folder, create a `States` folder
   - `Screenshots`
4. The final directory should look like the following image:
   - 

2.3 How to Set up ES-DE and the ROM Folders

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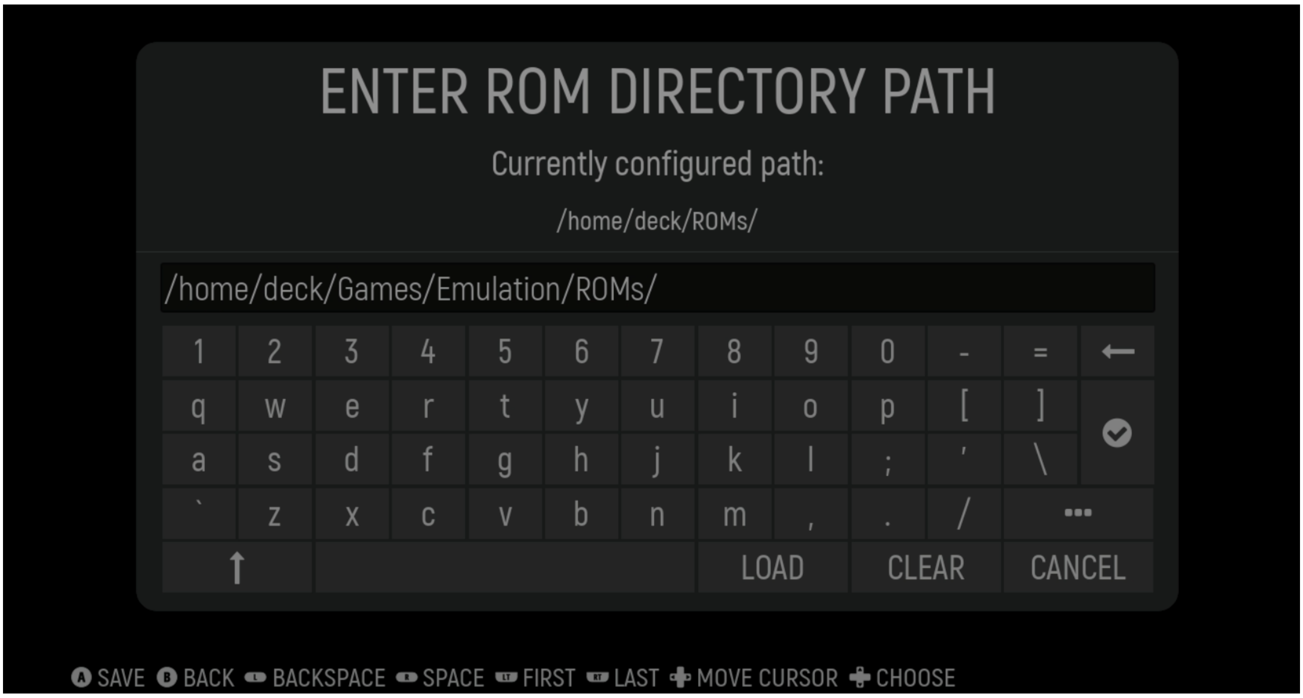
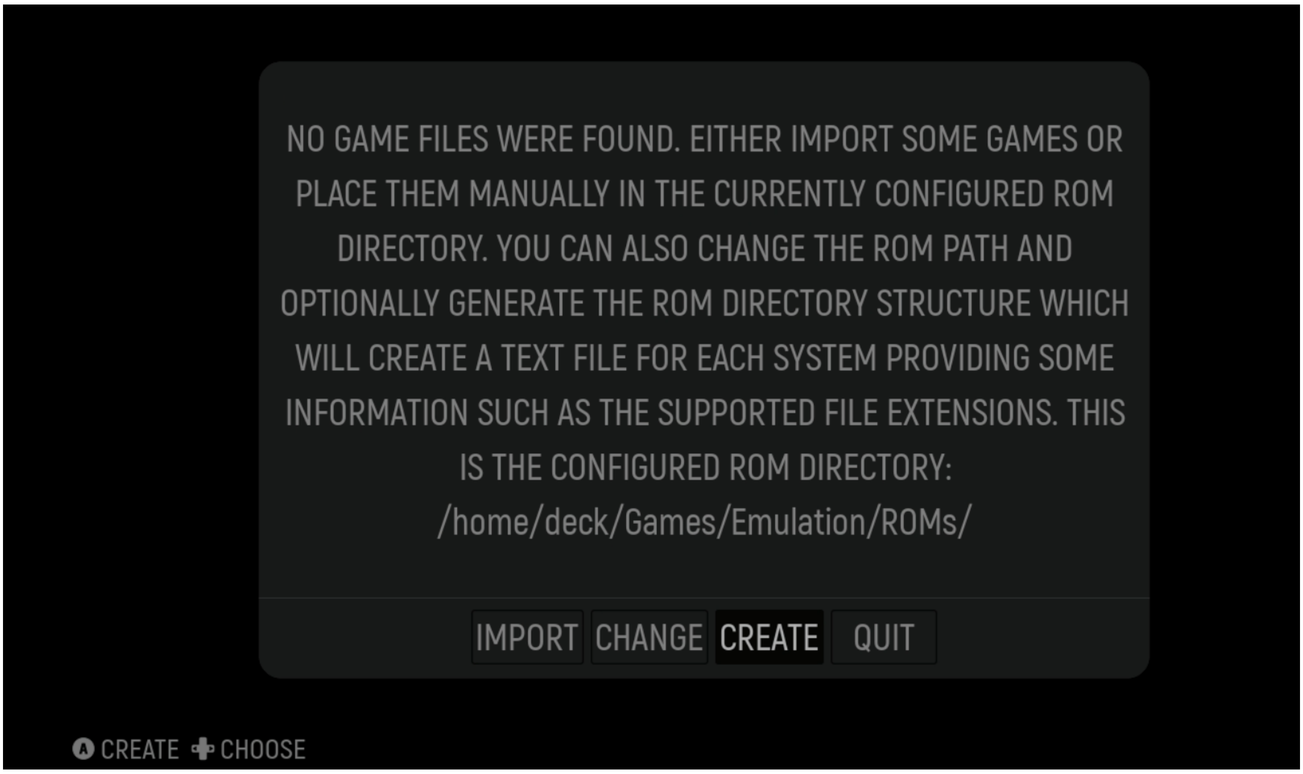
1. Open the ES-DE website, https://es-de.org/#Download
2. If on a Linux desktop, click the `Linux x64 AppImage` button. If on a Steam Deck, click the `Steam Deck AppImage`
3. Move the downloaded AppImage to `$HOME/Applications`.
4. Right click the application or file in question, click `Properties`, click `Permissions`, check `Is Executable`.
5. (Optional) For easier maintenance, rename the AppImage to `ES-DE.AppImage`
6. Double click the AppImage to run ES-DE
7. On the ES-DE onboarding screen, click `CHANGE` and update the path to `$HOME/Games/Emulation/ROMs`
    - 

8. Click the checkmark to confirm the path

9. On the onboarding screen, click `CREATE` to allow ES-DE to create the ROM folders in `$HOME/Games/Emulation/ROMs`
    - 

10. Once ES-DE has created the folders, click `QUIT`
11. ES-DE and the ROM folders are now configured

2.4 How to Set Up and Configure RetroArch

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2.4.1 How to Install RetroArch

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1. Open your distro's software manager.
    - On SteamOS and distros with KDE, you may use Discover, a "store" to install and manage Flatpaks.
2. Search for "RetroArch" and click "Install" on the top right of the software page For RetroArch
3. Open Konsole or a terminal of your choice
4. Type the following command and press enter after typing the command:
    - `flatpak override org.libretro.RetroArch --filesystem=host --user`
    - Alternatively, you may install and use Flatseal from your distro's software manager. Open Flatseal, locate the emulator on the left-hand side of the screen, scroll down to the `Filesystem` section and enable `All system files`
5. RetroArch will now be installed

2.4.2 How to Configure RetroArch

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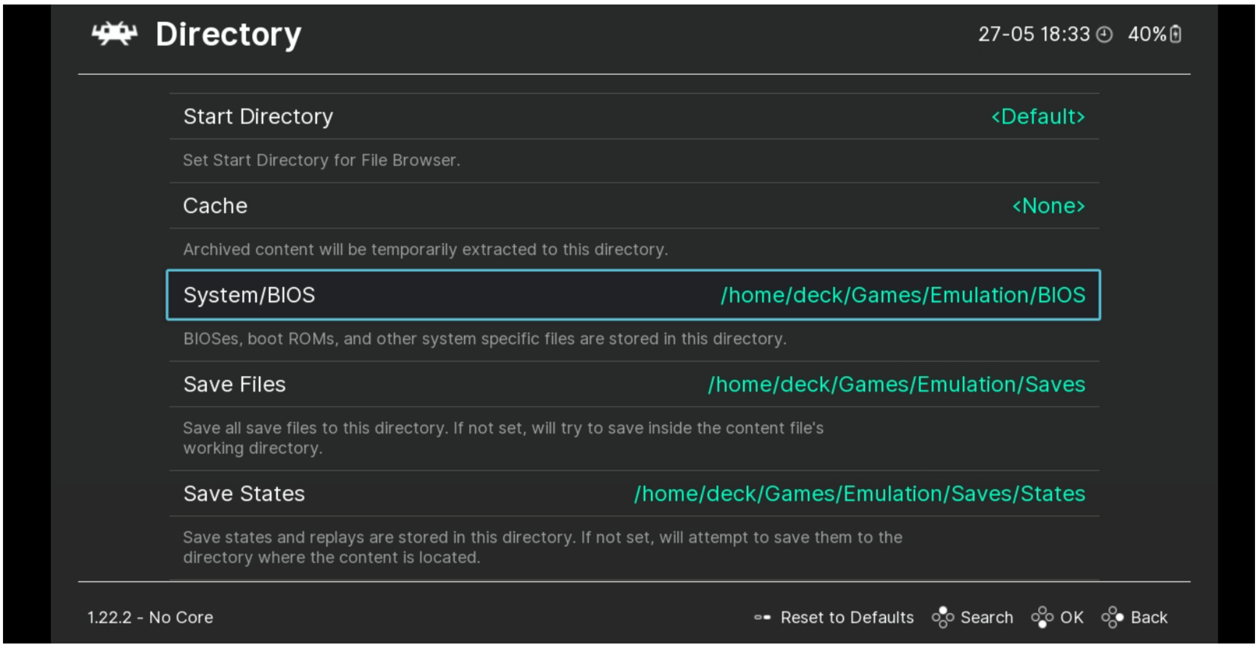
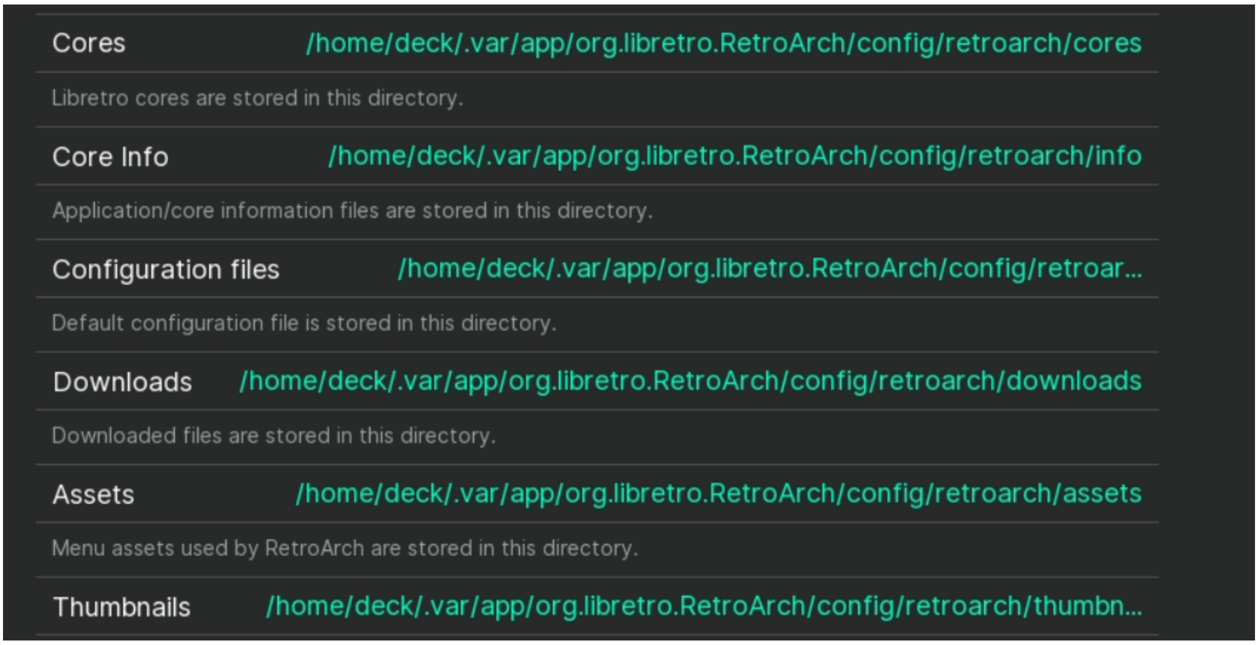
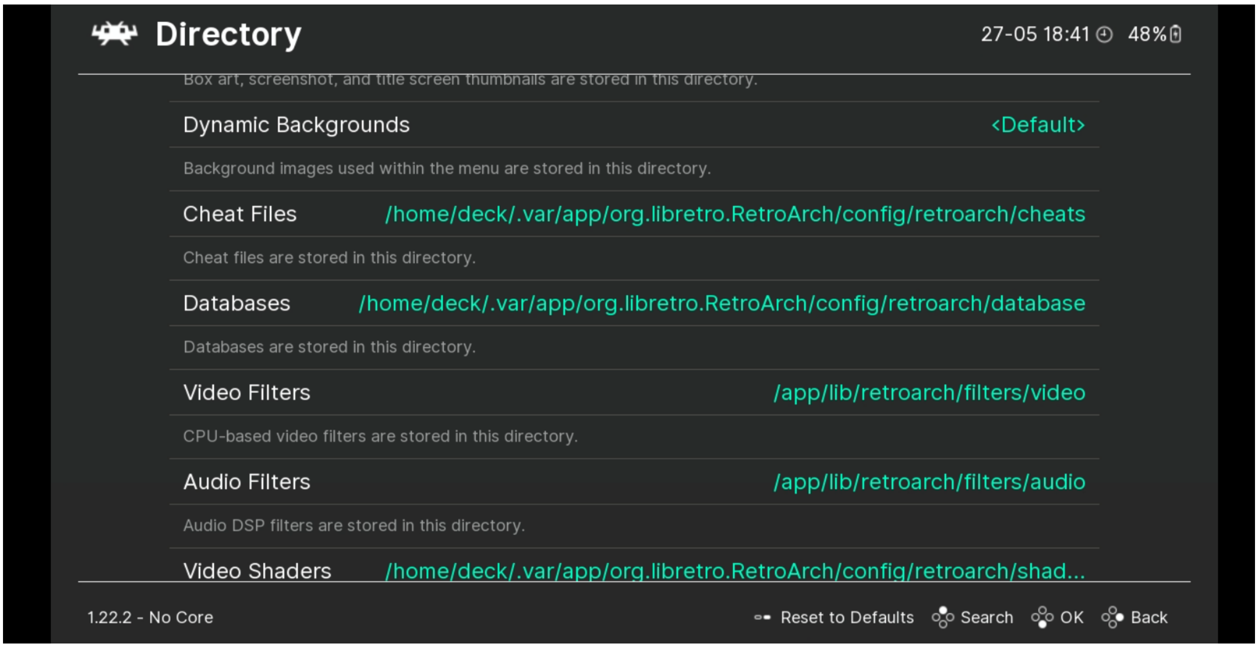
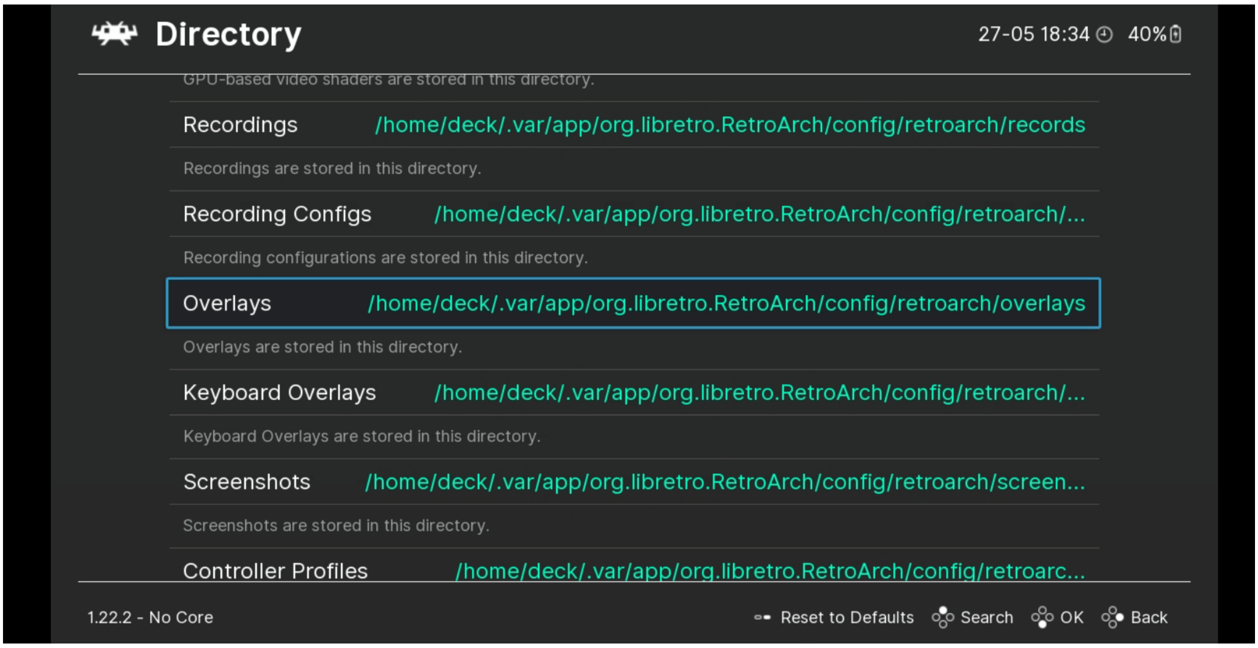
1. Launch RetroArch
2. Click `Settings` on the left-hand side of the screen
3. Scroll down to `Directory`

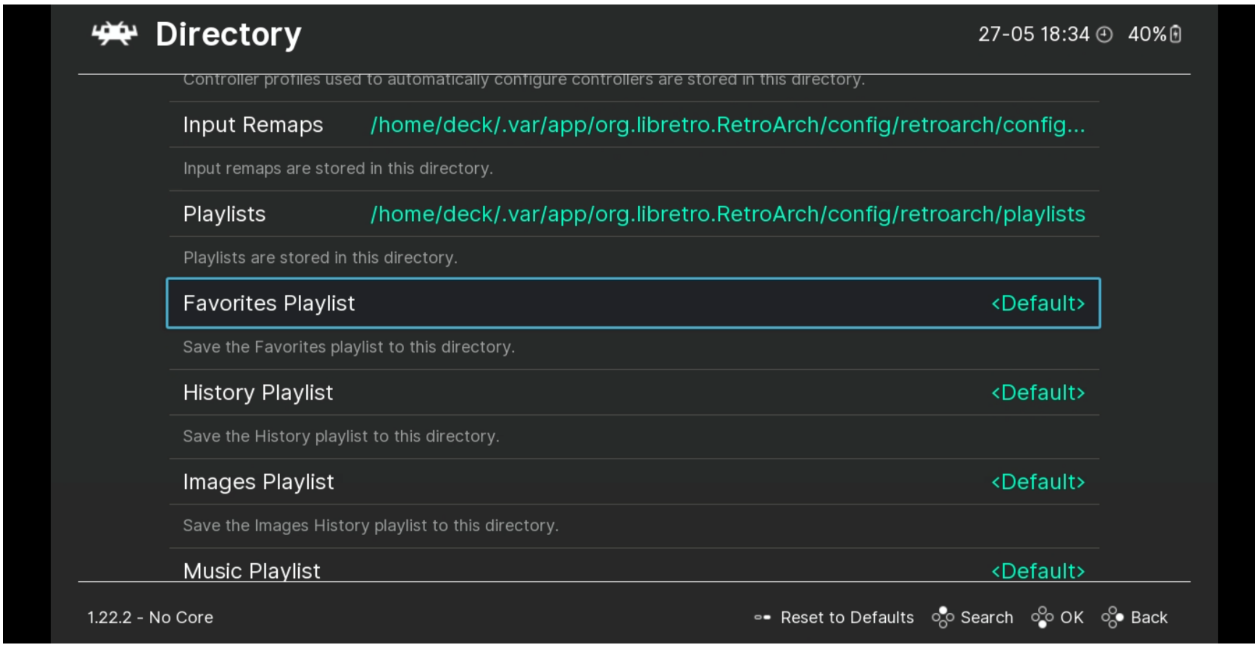
4. Use the following format:

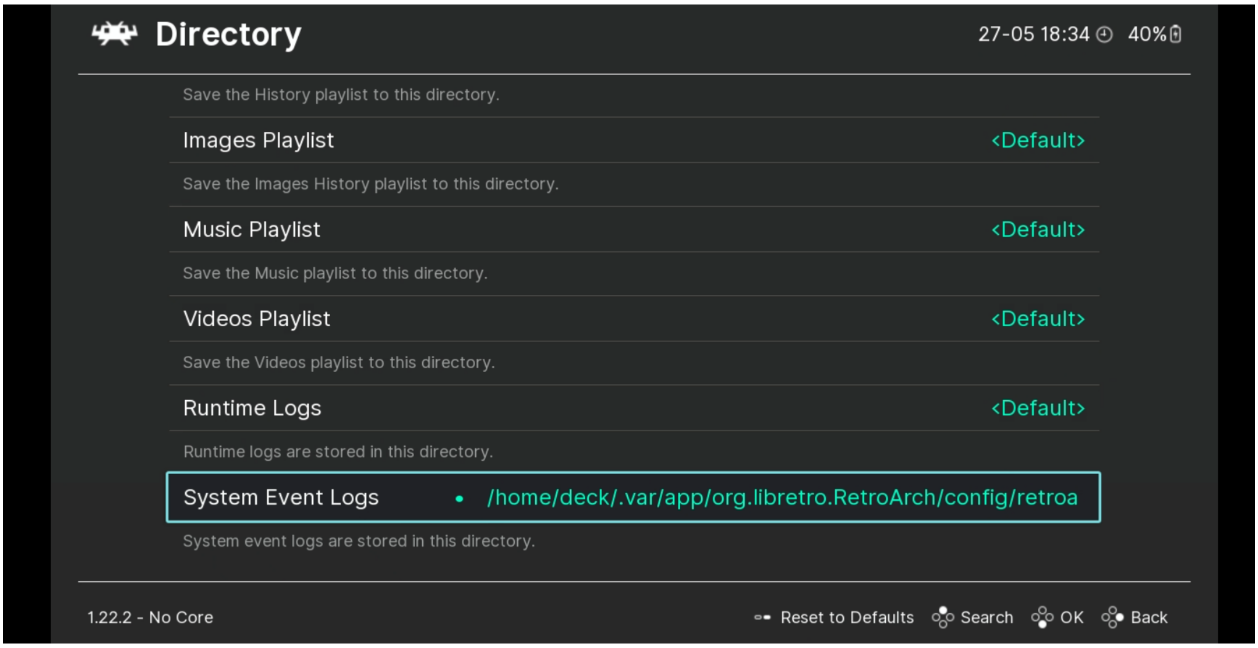
- For directores that start with `/app`, click `Parent Directory` until you see `$HOME/.var/app/org.libretro.RetroArch/config/retroarch`
- `System/BIOS`
    - `$HOME/Games/Emulation/BIOS`
- `Save Files`
    - `$HOME/Games/Emulation/Saves`
- `Save States`
    - `$HOME/Games/Emulation/Saves/States`
- `Core Info`
    - `$HOME/.var/app/org.libretro.RetroArch/config/retroarch/info`
- `Assets`
    - `$HOME/.var/app/org.libretro.RetroArch/config/retroarch/assets`
- `Cheats`
    - `$HOME/.var/app/org.libretro.RetroArch/config/retroarch/cheats`
- `Databases`
    - `$HOME/.var/app/org.libretro.RetroArch/config/retroarch/database`
- `Video Shaders`
    - `$HOME/.var/app/org.libretro.RetroArch/config/retroarch/shaders`
- `Overlays`
    - `$HOME/.var/app/org.libretro.RetroArch/config/retroarch/overlays`
- `Screenshots`
    - `$HOME/Games/Emulation/Screenshots`
- `Controller Profiles`
    - `$HOME/.var/app/org.libretro.RetroArch/config//retroarch/autoconfig`

• The final configuration screen should look similar to the image below:

- 
Directory
27-05 18:33
40%
Start Directory
<Default>
Set Start Directory for File Browser.
Cache
<None>
Archived content will be temporarily extracted to this directory.
System/BIOS
/home/deck/Games/Emulation/BIOS
BIOSes, boot ROMs, and other system specific files are stored in this directory.
Save Files
/home/deck/Games/Emulation/Saves
Save all save files to this directory. If not set, will try to save inside the content file's working directory.
Save States
/home/deck/Games/Emulation/Saves/States
Save states and replays are stored in this directory. If not set, will attempt to save them to the directory where the content is located.
1.22.2 - No Core
Reset to Defaults
Search
OK
Back
- 
Cores
/home/deck/.var/app/org.libretro.RetroArch/config/retroarch/cores
Libretro cores are stored in this directory.
Core Info
/home/deck/.var/app/org.libretro.RetroArch/config/retroarch/info
Application/core information files are stored in this directory.
Configuration files
/home/deck/.var/app/org.libretro.RetroArch/config/retroar...
Default configuration file is stored in this directory.
Downloads
/home/deck/.var/app/org.libretro.RetroArch/config/retroarch/downloads
Downloaded files are stored in this directory.
Assets
/home/deck/.var/app/org.libretro.RetroArch/config/retroarch/assets
Menu assets used by RetroArch are stored in this directory.
Thumbnails
/home/deck/.var/app/org.libretro.RetroArch/config/retroarch/thumbn...

- 
Directory
27-05 18:41 48%
Box art, screenshot, and title screen thumbnails are stored in this directory.
Dynamic Backgrounds <Default>
Background images used within the menu are stored in this directory.
Cheat Files /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/cheats
Cheat files are stored in this directory.
Databases /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/database
Databases are stored in this directory.
Video Filters /app/lib/retroarch/filters/video
CPU-based video filters are stored in this directory.
Audio Filters /app/lib/retroarch/filters/audio
Audio DSP filters are stored in this directory.
Video Shaders /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/shad...
1.22.2 - No Core
Reset to Defaults
Search
OK
Back
- 
Directory
27-05 18:34 40%
GPU-based video shaders are stored in this directory.
Recordings /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/records
Recordings are stored in this directory.
Recording Configs /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/...
Recording configurations are stored in this directory.
Overlays /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/overlays
Overlays are stored in this directory.
Keyboard Overlays /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/...
Keyboard Overlays are stored in this directory.
Screenshots /home/deck/.var/app/org.libretro.RetroArch/config/retroarch/screen...
Screenshots are stored in this directory.
Controller Profiles /home/deck/.var/app/org.libretro.RetroArch/config/retroarc...
1.22.2 - No Core
Reset to Defaults
Search
OK
Back

- 

- 

5. Back out of the `Directory` settings and click `Main Menu` on the left-hand side of the screen
6. Click `Configuration File`, click `Save Current Configuration`
7. Back out of the `Configuration File` screen and click `Online Updater`

8. Click the following options, one at a time, waiting for each to complete downloading
    - `Update Core Info Files`
    - `Update Assets`
    - `Update Controller Profiles`
    - `Update Cheats`
    - `Update Databases`
    - `Update Overlays`
    - `Update Cg Shaders`
    - `Update GLSL Shaders`
    - `Core System Files Downloader` > `PPSSPP.zip`
9. Optionally, click `Core Downloader` and select which cores you would like to use. If you would like to download in batch, skip to How to Download RetroArch Cores in Batch. Otherwise, RetroArch is now configured

2.4.3 How to Download RetroArch Cores in Batch

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1. Open the RetroArch buildbot website, https://buildbot.libretro.com/
2. click `Stable`, locate the latest version (look at the dates), click `linux`, `x86_64`
3. Download `RetroArch_cores.7z`
4. Right click `RetroArch_cores.7z`, click `Extract` > `Extract Here` which will extract it to a `RetroArch-Linux-x86-64` folder
5. Open the newly extracted folder and navigate through the subfolders until you locate the `cores` folder
6. Right click the `cores` folder, click `Copy`, navigate to `$HOME/.var/app/org.libretro.RetroArch/config/retroarch`
7. Right click anywhere in the folder, click `Paste`, click `Write Into`
8. All of the RetroArch cores are now downloaded

2.4.4 RetroArch Recommended Settings

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1. Launch RetroArch ⓘ
2. Click `Settings` on the left-hand side of the screen
3. Scroll down to `Video`
4. Click `Output`, `Video`, and select `vulkan`
5. Back out to the `Video` settings, click `Fullscreen Mode`, and click `Fullscreen Display` to enable it
6. Back out to the `Settings` menu, click `Audio`, click `Output`, `Audio`, select `Pulsewire`
7. Back out to the `Audio` menu, click `Microphone`, `Microphone`, and select `sdl2`
8. Back out to the `Settings` menu, click `Input`, `Controller`, and select `sdl2`
9. Back out to the `Input` menu, click `Controller`, and select `sdl2`
10. Back out to the `Settings` menu, click `User Interface`, `File Browser`, `Start Directory`, and set it to `$HOME/Games/Emulation/ROMs`
11. Back out to the `User Interface` menu and enable `Remember Last Used Start Directory`
12. Back out to the `Settings` menu, click `Main Menu` on the left-hand side of the screen, click `Configuration File`, click `Save Current Configuration`

2.4.5 How to Configure Hotkeys

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1. Launch RetroArch ⓘ
2. Click `Settings` on the left-hand side of the screen
3. Scroll down to `Input`
4. Click `Hotkeys`
5. Click `Hotkey` enable and press `Select` on your controller
6. Use the following format:
    - `Menu Toggle (Controller Combo)`
        - `L3 + R3`
    - `Menu Toggle`
        - `R3`
    - `Quit (Controller Combo)`
        - `Start + Select`
    - `Reset`
        - `L3`
    - `Fast Forward (Toggle)`
        - `R2`
    - `Rewind`
        - `L2`
    - `Pause`
        - `A`
    - `Save State`
        - `R1`
    - `Load State`
        - `L1`
    - `Next Save State Slot`
        - `DPad Right`
    - `Previous Save State Slot`
        - `DPad Left`
    - `Next Disc`
        - `Y`
    - `Take Screenshot`
        - `B`
7. Back out to the `Settings` menu, click `Main Menu` on the left-hand side of the screen, click `Configuration File`, click `Save Current Configuration`

2.4.6 How to Configure Shaders

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See https://retrogamecorps.com/2022/02/28/retroarch-starter-guide/#Shaders and https://retrogamecorps.com/2024/09/01/guide-shaders-and-overlays-on-retro-handhelds/ for an in-depth guide.

2.4.7 How to Configure RetroAchievements

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1. Launch RetroArch ⓘ
2. Click `Settings` on the left-hand side of the screen
3. Scroll down to `Achievements`
4. Click `Achievements` to toggle it to `On`
5. Enter your `Username` and `Password` in the respective fields
6. Back out to the `Settings` menu, click `Main Menu` on the left-hand side of the screen, click `Configuration File`, click `Save Current Configuration`

2.5 How to Play a Game in ES-DE Using RetroArch

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This section will use Apotris, a homebrew Tetris game for the Game Boy Advance as an example.

1. Open https://akouzoukos.com/apotris/downloads on a browser of your choice
2. Click `GameBoy Advance` under `Downloads`
3. Extract the newly downloaded `.zip` file
4. In the newly extracted folder, move `Apotris.gba` to `$HOME/Games/Emulation/ROMs/gba`
5. Open ES-DE
6. Apotris should now appear under the `GameBoy Advance` collection
7. On the `Apotris` entry, click `Select`, `Edit This Game's Metadata`, `Alternative Emulator` and confirm `mGBA [SYSTEM-WIDE]` is selected
8. Save the changes and launch the `ROM` by selecting it, it should now open in RetroArch

2.6 How to Set Up Steam ROM Manager

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2.6.1 How to Download Steam ROM Manager

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1. Open the Steam ROM Manager GitHub releases page, https://github.com/SteamGridDB/steam-rom-manager/releases on a browser of your choice ⓘ
2. Download the latest `Steam-ROM-Manager-#.#.##.AppImage` file
   - The `#.#.##` refer to the version number, this may vary depending on when you download Steam ROM Manager
3. Move the newly downloaded AppImage to the `$HOME/Applications` folder
   - $HOME/Applications refers to the Applications folder in your home directory. On a typical linux distro, the full path will be $HOME/$USER/Applications. On a Steam Deck, the full path will be `/home/deck/Applications`. When navigating using the file manager, the file manager may not display the username (the username being `deck` on a Steam Deck) visually in the full file path. The Applications folder is not created by default. If you have not created one already, do so before following the steps in this section or on this page.
4. Right click the application or file in question, click `Properties`, click `Permissions`, check `Is Executable`.
5. (Optional) For easier maintenance, rename the AppImage to `Steam-ROM-Manager.AppImage`
6. To launch Steam ROM Manager, double click the AppImage

2.6.2 How to Configure Steam ROM Manager

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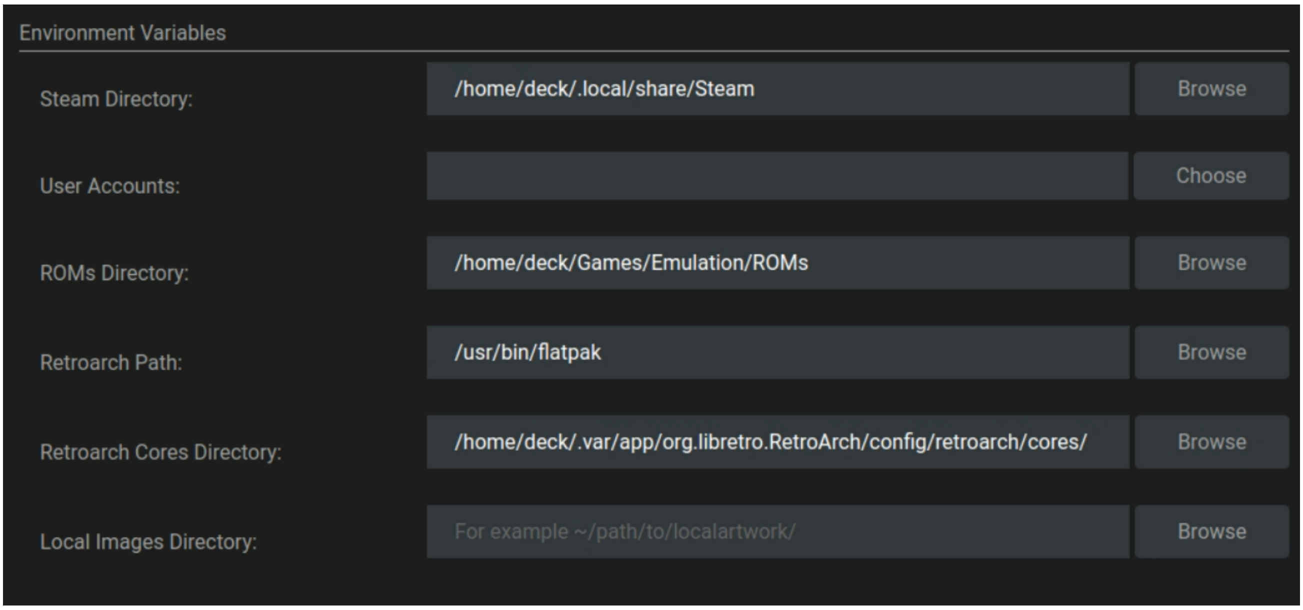
1. Launch Steam ROM Manager
2. On the `Welcome to SRM` screen, select your Steam directory
    - If you are on a Steam Deck, the path will be `/home/deck/.local/share/steam`
    - To show the hidden folder in Steam ROM Manager's file browser, right click anywhere in the blank space, click `Show Hidden Folders`
3. On the next screen, select your Steam account
4. On the left-hand side of the screen, click `Settings`
5. Enable `Auto kill Steam` and `Auto restart Steam`
6. Scroll down to `Environment Variables`
7. In the `ROMs Directory` field, write `$HOME/Games/Emulation/ROMs` or the ROM path where your ROMs are located
8. In the `Retroarch Path` field, write `/usr/bin/flatpak`
9. In the `Retroarch Cores Directory` field, write `$HOME/.var/app/org.libretro.RetroArch/config/retroarch/cores`
    - To show the hidden folder in Steam ROM Manager's file browser, click the `Menu` in the top right corner, click `Show Hidden Folders`
10. The final screen should look similar to the image below (with the `User Accounts` field populated as well)
    - 

11. Steam ROM Manager is now configured, you may now proceed to How to Set up a RetroArch Core Parser

2.6.3 How to Set up a RetroArch Core Parser

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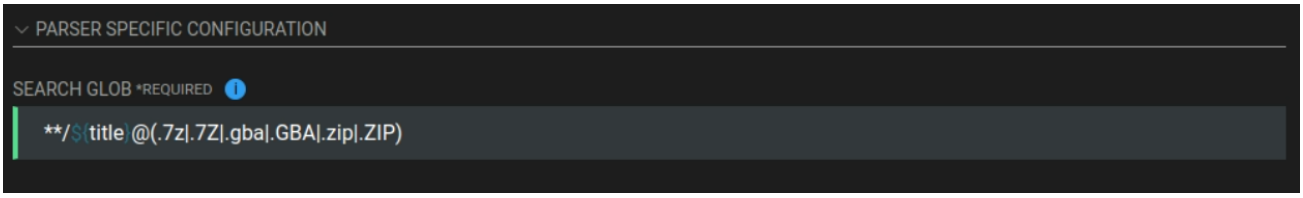
This section will use Apotris, a homebrew Tetris game for the Game Boy Advance as an example.

1. Open https://akouzoukos.com/apotris/downloads on a browser of your choice
2. Click `GameBoy Advance` under `Downloads`
3. Extract the newly downloaded `.zip` file
4. In the newly extracted folder, move `Apotris.gba` to `$HOME/Games/Emulation/ROMs/gba`
5. Launch Steam ROM Manager
6. Click `Create Parser` on the left-hand side of the screen
7. Search for `mgba` under `Community Presets` and select `Nintendo Game Boy Advance - RetroArch(Flatpak) - mGBA`
8. In the `ROMS Directory` field, write the following:
    - `${romsdirglobal}/gba`
    - For creating any other parser, replace `gba` with the respective ROM folder
9. (Optional) If you use subfolders in your ROM folders, in the `Search Glob` field, add the following to the beginning of the glob:
    - `**/`
    - 

    - The general format to use subfolders is the following:
        - `**/${title}@(.7z|.7Z|.gba|.GBA|.zip|.ZIP)`
        - Replace the file formats with your file formats, keeping in mind that it is case sensitive
10. Click `Test` to validate that the parser can identify and locate your ROMs
11. If the `Test` succeeds, click `Save`
12. Ensure the newly created parser is enabled with a green indicator on the left-hand side of the screen
13. Click `Add Games`
14. Click `Parse`
15. Select your artwork, when you are ready, click `Save to Steam`

2.7 How to Set up Freegosy

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> **Info**
>
> Freegosy is a front-end with ROMM integrated. The primary benefit of ROMM is access to your ROM library on a separate server such as a NAS for example. When using Freegosy, you are able to download ROMs directly from your own collection and launch them immediately from Freegosy. Alternatively with the ROMs downloaded, you may then use Steam ROM Manager to parse and add them to Steam or launch them from ES-DE.

2.7.1 How to Download Freegosy

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1. Open the Freegosy GitHub releases page, https://github.com/abduznik/Freegosy/releases on a browser of your choice
2. Download the latest `Freegosy-x86_64.AppImage` file
3. Move the newly downloaded AppImage to the `$HOME/Applications` folder
    - $HOME/Applications refers to the Applications folder in your home directory. On a typical linux distro, the full path will be $HOME/$USER/Applications. On a Steam Deck, the full path will be `/home/deck/Applications`. When navigating using the file manager, the file manager may not display the username (the username being `deck` on a Steam Deck) visually in the full file path. The Applications folder is not created by default. If you have not created one already, do so before following the steps in this section or on this page.
4. Right click the application or file in question, click `Properties`, click `Permissions`, check `Is Executable`.
5. (Optional) For easier maintenance, rename the AppImage to `Freegosy.AppImage`
6. To launch Freegosy, double click the AppImage

At this time, Freegosy can be used to manage and download ROMs **from your own** collection. Launching games is still in an early phase and it is generally recommended to use ES-DE or Steam ROM Manager.

2.8 Steam Deck: How to Add RetroArch to Steam and Play RetroArch in Game Mode

If on a Steam Deck or SteamOS, right click RetroArch in the applications launcher and click `Add to Steam`. When you are in Game Mode, RetroArch will appear in the `Non-Steam Games` tab and can be launched directly from Game Mode.

2.9 Final Words and Tips

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Make sure you update the various items in the `Online Updater` in RetroArch on major RetroArch updates to keep everything up to date. For steps, see Step 9 in How to Configure RetroArch.

May 29, 2026

November 16, 2025

3. The Cheat Sheet is Your Tool to Success.

3.1 Cheat Sheet Table of Contents

3.2 Introduction

3.2.1 ROMs

This cheat sheet assumes you are using ES-DE's file directory for ROMs. For a full list of these folders, see below.

Folder Names

3do: 3DO Interactive Multiplayer
adam: Coleco Adam
ags: Adventure Game Studio Game Engine
amiga: Commodore Amiga
amiga1200: Commodore Amiga 1200
amiga600: Commodore Amiga 600
amigacd32: Commodore Amiga CD32
amstradcpc: Amstrad CPC
apple2: Apple II
apple2gs: Apple IIGS
arcade: Arcade
arcadia: Emerson Arcadia 2001
archimedes: Acorn Archimedes
arduboy: Arduboy Miniature Game System
astrocde: Bally Astrocade
atari2600: Atari 2600
atari5200: Atari 5200
atari7800: Atari 7800 ProSystem
atari800: Atari 800
atarijaguar: Atari Jaguar
atarijaguarcd: Atari Jaguar CD
atarilynx: Atari Lynx
atarist: Atari ST
atarixe: Atari XE
atomiswave: Sammy Corporation Atomiswave
bbcmicro: Acorn Computers BBC Micro
c64: Commodore 64
cdimono1: Philips CD-i
cdtv: Commodore CDTV
chailove: ChaiLove Game Engine
channelf: Fairchild Channel F
coco: Tandy Color Computer
colecovision: Coleco ColecoVision
consolearcade: Console Arcade Systems
cps: Capcom Play System
cps1: Capcom Play System I
cps2: Capcom Play System II
cps3: Capcom Play System III
crvision: VTech CreatiVision
daphne: Daphne Arcade LaserDisc Emulator
desktop: Desktop Applications
doom: Doom
dos: DOS (PC)
dragon32: Dragon Data Dragon 32
dreamcast: Sega Dreamcast
easyrpg: EasyRPG Game Engine
electron: Acorn Electron
emulators: Emulators
epic: Epic Games Store
famicom: Nintendo Family Computer
fba: FinalBurn Alpha
fbneo: FinalBurn Neo
fds: Nintendo Famicom Disk System
flash: Adobe Flash
fm7: Fujitsu FM-7
fmtowns: Fujitsu FM Towns
gamate: Bit Corporation Gamate

gameandwatch: Nintendo Game and Watch
gamecom: Tiger Electronics Game.com
gamegear: Sega Game Gear
gb: Nintendo Game Boy
gba: Nintendo Game Boy Advance
gbc: Nintendo Game Boy Color
gc: Nintendo GameCube
genesis: Sega Genesis
gmaster: Hartung Game Master
gx4000: Amstrad GX4000
intellivision: Mattel Electronics Intellivision
j2me: Java 2 Micro Edition (J2ME)
kodi: Kodi Home Theatre Software
laserdisc: LaserDisc Games
lcdgames: LCD Handheld Games
lindbergh (custom system): Sega Lindbergh
lowresnx: LowRes NX Fantasy Console
lutris: Lutris Open Gaming Platform
lutro: Lutro Game Engine
macintosh: Apple Macintosh
mame: Multiple Arcade Machine Emulator
mame-advmame: AdvanceMAME
mastersystem: Sega Master System
megacd: Sega Mega-CD
megacdjp: Sega Mega-CD
megadrive: Sega Mega Drive
megadrivejp: Sega Mega Drive
megaduck: Creatronic Mega Duck
mess: Multi Emulator Super System
model2: Sega Model 2
model3: Sega Model 3
moto: Thomson MO/TO Series
msx: MSX
msx1: MSX1
msx2: MSX2
msxturbor: MSX Turbo R
mugen: M.U.G.E.N Game Engine
multivision: Othello Multivision
n3ds: Nintendo 3DS
n64: Nintendo 64
n64dd: Nintendo 64DD
naomi: Sega NAOMI
naomi2: Sega NAOMI 2
naomigd: Sega NAOMI GD-ROM
nds: Nintendo DS
neogeo: SNK Neo Geo
neogeocd: SNK Neo Geo CD
neogeocdjp: SNK Neo Geo CD
nes: Nintendo Entertainment System
ngage: Nokia N-Gage
ngp: SNK Neo Geo Pocket
ngpc: SNK Neo Geo Pocket Color
odyssey2: Magnavox Odyssey 2
openbor: OpenBOR Game Engine
oric: Tangerine Computer Systems Oric
palm: Palm OS
pc: IBM PC
pc88: NEC PC-8800 Series
pc98: NEC PC-9800 Series

pcarcade: PC Arcade Systems
pcengine: NEC PC Engine
pcenginecd: NEC PC Engine CD
pcfx: NEC PC-FX
pico8: PICO-8 Fantasy Console
plus4: Commodore Plus/4
pokemini: Nintendo Pokémon Mini
ports: Ports
ps2: Sony PlayStation 2
ps3: Sony PlayStation 3
psp: Sony PlayStation Portable
psvita: Sony PlayStation Vita
psx: Sony PlayStation
pv1000: Casio PV-1000
quake: Quake
samcoupe: MGT SAM Coupé
satellaview: Nintendo Satellaview
saturn: Sega Saturn
saturnjp: Sega Saturn
scummvm: ScummVM Game Engine
scv: Epoch Super Cassette Vision
sega32x: Sega Mega Drive 32X
sega32xjp: Sega Super 32X
sega32xna: Sega Genesis 32X
segacd: Sega CD
sfc: Nintendo SFC (Super Famicom)
sg-1000: Sega SG-1000
sgb: Nintendo Super Game Boy
snes: Nintendo SNES (Super Nintendo)
snesna: Nintendo SNES (Super Nintendo)
solarus: Solarus Game Engine
spectravideo: Spectravideo
steam: Valve Steam
stv: Sega Titan Video Game System
sufami: Bandai SuFami Turbo
supergrafx: NEC SuperGrafx
supervision: Watara Supervision
supracan: Funtech Super A'Can
switch: Nintendo Switch
symbian: Symbian
tanodragon: Tano Dragon
tg-cd: NEC TurboGrafx-CD
tg16: NEC TurboGrafx-16
ti99: Texas Instruments TI-99
tic80: TIC-80 Fantasy Computer
to8: Thomson TO8
triforce: Namco-Sega-Nintendo Triforce
trs-80: Tandy TRS-80
type-x: Taito Type X
uzebox: Uzebox Open Source Console
vectrex: GCE Vectrex
vic20: Commodore VIC-20
videopac: Philips Videopac G7000
virtualboy: Nintendo Virtual Boy
vpinball: Visual Pinball
vsmile: VTech V.Smile
wasm4: WASM-4 Fantasy Console
wii: Nintendo Wii
wiiu: Nintendo Wii U

windows: Microsoft Windows
windows3x: Microsoft Windows 3.x
windows9x: Microsoft Windows 9x
wonderswan: Bandai WonderSwan
wonderswancolor: Bandai WonderSwan Color
x1: Sharp X1
x68000: Sharp X68000
xbox: Microsoft Xbox
xbox360: Microsoft Xbox 360
zmachine: Infocom Z-machine
zx81: Sinclair ZX81
zxnext: Sinclair ZX Spectrum Next
zxspectrum: Sinclair ZX Spectrum

3.2.2 BIOS

This cheat sheet assumes you are using a `bios` or `BIOS` folder for your BIOS. Paths listed will be relative and will not assume a hard-coded path. When reading `bios/BIOSNAME.bin`, you may replace the `bios` folder with the path to your own BIOS folder.

3.3 Cheat Sheets

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3.3.1 Arcade and MAME Related Emulation Cheat Sheet

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Console/System	ROM Folder Name	Emulator (Click for Link)	File Types	BIOS Files
Final Burn Neo	fbneo	RetroArch FinalBurn Neo	.7z .zip	BIOS will be searched through 3 folders: • `roms/arcade/` • `bios/mame/` • `bios/` •EmulationStation-DE will ONLY work with the BIOS in `roms/arcade/` folder
MAME 2003 Plus	mame2003	RetroArch MAME 2003 Plus	.zip	Not Required
MAME 2010	mame2010	RetroArch MAME 2010	.zip	Not Required
MAME Current	arcade	RetroArch MAME Current	.zip	Not Required
MAME (Standalone)	arcade	**MAME (Standalone)**	.chd .zip	BIOS will be searched through 3 folders: • `roms/arcade/` • `bios/mame/` • `bios/` •EmulationStation-DE will ONLY work with the BIOS in `roms/arcade/` folder
Philips CD-i	cdimono1	RetroArch SAME CDi	.chd .iso	Create `same_cdi/bios` folders in `bios` Place `cdibios.zip` in `bios/same_cdi/bios` May substitute `cdibios.zip` with `cdimono.zip` or `cdimono2.zip`
Philips CD-i (Standalone)	cdimono1	**MAME (Standalone)**	.chd .iso	Place `cdimono1.zip` in `roms/cdimono`
Tiger Electronics Game.com (Standalone)	gamecom	**MAME (Standalone)**	.zip	Place `gamecom.zip` in `roms/gamecom`
VTech V.Smile (Standalone)	vsmile	**MAME (Standalone)**	.zip	Place `vsmile.zip` in `roms/vsmile`

- **Final Burn Neo**
- **MAME 2003**
- **MAME 2010**
- **MAME Current**
- **MAME (Standalone)**
- **Philips CD-i**
- **Tiger Electronics Game.com**
- **VTech V.Smile**

3.3.2 Atari Cheat Sheet

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Console/System	ROM Folder Name	Emulator (Click for Link)	File Types	BIOS Files
Atari 2600	atari2600	RetroArch Stella	.7z .a26 .bin .zip	Not Required
Atari Jaguar	atarijaguar	BigPEmu	.abs .cdi .cof .cue .j64 .jag .rom .zip	Not Required
Atari Jaguar	atarijaguarcd	BigPEmu	.abs .cdi .cof .cue .j64 .jag .rom .zip	Not Required
Atari Lynx	lynx	RetroArch Beetle Lynx	.7z .bin .lnx .zip	`lynxboot.img`

3.3.3 Bandai Cheat Sheet

Console/System	ROM Folder Name	Emulator (Click for Link)	File Types	BIOS Files
Bandai Wonderswan	wonderswan	RetroArch Beetle Cygne	.7z .pc2 .ws .wsc .zip	Not Required
Bandai Wonderswan Color	wonderswancolor	RetroArch Beetle Cygne	.7z .pc2 .ws .wsc .zip	Not Required

3.3.4 Game Engine Recreations Cheat Sheet

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Console/System	ROM Folder Name	Emulator (Click for link)	File Types	BIOS Files
DooM	doom	RetroArch PrBoom	.7z .iwad .pwad .wad .zip	`prboom.wad`
EasyRPG	easyrpg	RetroArch EasyRPG	.easyrpg .ldb	Not Required
Pico-8	pico8	RetroArch Pico-8	.7z .p8 .png .zip	Not Required
ScummVM (Standalone)	scummvm	**ScummVM (Standalone)**	Varies	Not Required

- **DooM**
- **EasyRPG**
- **Pico-8**
- **ScummVM**

3.3.5 Mattel Cheat Sheet

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Console/System	ROM Folder Name	Emulator (Click for link)	File Types	BIOS Files
Mattel Electronics Intellivison	intellivision	RetroArch FreeIntv	.7z .bin .int .rom .zip	Not Required

- **Mattel Electronics Intellivison**

3.3.6 Microsoft Cheat Sheet

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Console/System	ROM Folder Name	Emulator (Click for Link)	File Types	BIOS Files
Microsoft Xbox (Standalone)	xbox	**Xemu (Standalone)**	.iso (xiso formatted)	MCPX Boot ROM: `mcpx_1.0.bin` Flash ROM Image (COMPLEX 4627 BIOS): `Complex_4627v1.03.bin`
Microsoft Xbox 360 (Standalone)	`xbox360/roms`	**Xenia (Standalone)**	.iso .zar	Not Required
Microsoft Xbox 360 Live Arcade (Standalone)	`xbox360/roms/xbla`	**Xenia (Standalone)**	No file extension .zar	Not Required

- **Microsoft Xbox**
- **Microsoft Xbox 360**
- **Microsoft MSX 1**
- **Microsoft MSX 2**

3.3.7 NEC Cheat Sheet

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Console/System	ROM Folder Name	Emulator (Click for Link)	File Types	BIOS Files
NEC TurboGrafx-16 / PC Engine	`tg16` or `pcengine`	RetroArch Beetle PCE	.bin .ccd `.cue and .bin` .chd .img .iso .pce	`syscard1.pce` `syscard2.pce` `syscard3.pce` `gexpress.pce`
NEC TurboGrafx-16 CD / PC Engine CD	`tg-cd` or `pcenginecd`	RetroArch Beetle PCE	.bin .ccd `.cue and .bin` .chd .img .iso .pce	`syscard1.pce` `syscard2.pce` `syscard3.pce` `gexpress.pce`

- **NEC TurboGrafx-16 / PC Engine**
- **TurboGrafx-16 / PC Engine CD**
- **NEC SuperGrafx**

3.3.8 Nintendo Cheat Sheet

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System	ROM Folder Name	Emulator (Click for Link)	File Format	BIOS Files
Nintendo 3DS (Standalone)	`n3ds`	**Azahar (Standalone)**	.app .axf cci .cxi .elf .cia is incompatible with ES-DE and Steam ROM Manager .cia can only be used if installed and played through Citra's GUI	Place `aes_keys.tx` encrypted RO `bios/citra/`
Nintendo 64 (Standalone)	n64	**Rosalie's Mupen GUI (Standalone)**	.7z `.bin` .n64 .ndd .u1 .v64 .z64 .zip	Not Required

Nintendo 64DD (Standalone)	`n64` or `n64dd`	**Rosalie's Mupen GUI (Standalone)**	.7z `.bin` .n64 .ndd .u1 .v64 .z64 .zip	Nintendo 64D BIOS:
				`64DD_IPL_US`
				`64DD_IPL_JP`
				`64DD_IPL_DE`

Nintendo 64	n64	RetroArch Mupen64plus-Next	.7z `.bin` .n64 .ndd .u1 .v64 .z64 .zip	Not Required

Nintendo DS (Standalone)	nds	**melonDS (Standalone)**	.nds	**Optional:**
				Nintendo DS
				`bios7.bin`
				`bios9.bin`
				`firmware.bi`

Nintendo DSiWare (Standalone)	nds	**melonDS (Standalone)**	.app
			How to Set Up DSIWare
				Nintendo DSI
				`dsi_bios7.b`
				`dsi_bios9.b`
				`dsi_firmwar`
				`dsi_nand.bi`

Nintendo DS	nds	RetroArch melonDS (Legacy Core)	.7z .nds .zip	Nintendo DS
				`bios7.bin`
				`bios9.bin`
				`firmware.bi`

Nintendo DS	nds	RetroArch melonDS DS	.7z .nds .zip	**Optional:**
				Nintendo DS
				`bios7.bin`
				`bios9.bin`
				`firmware.bi`

Nintendo Game Boy (Standalone)	gb	**mGBA (Standalone)**	.7z .dmg .gb .zip	Not Required
Nintendo Game Boy	gb	RetroArch SameBoy	.7z .dmg .gb .zip	Not Required
Nintendo Game Boy	gb	RetroArch Gambatte	.7z .dmg .gb .zip	Not Required
Nintendo Game Boy Advance (Standalone)	gba	**mGBA (Standalone)**	.7z .gba .zip	Not Required
Nintendo Game Boy Advance	gba	RetroArch mGBA	.7z .gba .zip	Not Required
Nintendo Game Boy Color (Standalone)	gbc	**mGBA (Standalone)**	.7z .dmg .gb .gbc .zip	Not Required

System	ROM Folder Name	Emulator (Click for Link)	File Format	BIOS Files
Nintendo Game Boy Color	gbc	RetroArch SameBoy	.7z .dmg .gb .gbc .zip	Not Required
Nintendo Game Boy Color	gbc	RetroArch Gambatte	.7z .dmg .gb .gbc .zip	Not Required
Nintendo GameCube (Standalone)	`gc`	**Dolphin (Standalone)**	.ciso .dol .elf .gcm .gcz .iso .m3u .nkit .rvz .wad .wbfs .wia	**Optional:** Gamecube BI `IPL.bin` in t correct regio
Nintendo NES / Famicom	`nes` or `famicom`	RetroArch Mesen	.7z .fds .nes .unf .unif .zip	FDS (Famico System) gam require a BIO `disksys.rom` `bios`
Nintendo NES / Famicom	`nes` or `famicom`	RetroArch Nestopia	.7z .fds .nes .unf .unif .zip	FDS (Famico System) gam require a BIO `disksys.rom` `bios`
Super Nintendo	`snes` or `snesna`	RetroArch Snes9x	.7z .bs .fig .sfc .smc .swx .zip	Not Required
Super Nintendo Widescreen	`sneshd`	RetroArch bsnes hd beta	.7z .bs .fig .sfc .smc .swx .zip	Not Required
Nintendo PrimeHack (Standalone)	primehacks	**PrimeHack (Metroid Prime) (Standalone)**	.ciso .dol .elf .gcm .gcz .iso .json .nkit .rvz .wad .wbfs .wia	Not Required
Nintendo Switch (Standalone)	switch	**Ryujinx (Standalone)**	.kp .nca .nro .nso .nsp .nsz .xci	• Place `prod` in: `bios/ryuj` `keys` • Install firmw through Ryuji • **Both** firmw keys are **requ** Ryujinx
Nintendo Switch (Standalone)	switch	**Yuzu (Standalone)**	.kp .nca .nro .nso .nsp .nsz .xci	• Place `prod` in: `bios/yuzu` • Place `Firm` files in: `bios` `firmware`
Nintendo Virtual Boy	virtualboy	RetroArch Beetle VB	.7z .bin .vb .vboy .zip	Not Required
Nintendo Wii (Standalone)	wii	**Dolphin (Standalone)**	.ciso .dol .elf .gcm .gcz .iso .json .m3u .nkit .rvz .wad .wbfs .wia	Not Required
Nintendo Wii U (Standalone)	`wiiu/roms` ⓘ	**Cemu (Standalone) (Native)**	.elf .rpx .wad .wua .wud .wux	Place `keys.t` encrypted RO `/home/deck/` `share/Cemu/`

3.3.9 Panasonic Cheat Sheet

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Console/System	ROM Folder Name	Emulator (Click for Link)	File Types	BIOS Files
Panasonic 3DO	3do	RetroArch Opera	.chd `.cue and bin` .iso	`panafz1.bin`

- **Panasonic 3DO**

3.3.10 Personal Computers Cheat Sheet

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System	ROM Folder Name	Emulator	File Format
Amiga	amiga	RetroArch PUAE	.7z .adf .adz .ccd .chd `.cue and .bin` .dms .fdi .hdf .hdz .info .ipf .iso .lha .m3u .mds
Amiga 600	amiga600	RetroArch PUAE	.7z .adf .adz .ccd .chd `.cue and .bin` .dms .fdi .hdf .hdz .info .ipf .iso .lha .m3u .mds
Amiga 1200	amiga1200	RetroArch PUAE	.7z .adf .adz .ccd .chd `.cue and .bin` .dms .fdi .hdf .hdz .info .ipf .iso .lha .m3u .mds
Amiga CD32	amigacd32	RetroArch PUAE	.7z .adf .adz .ccd .chd `.cue and .bin` .dms .fdi .hdf .hdz .info .ipf .iso .lha .m3u .mds
Amstrad CPC	amstradcpc	RetroArch Caprice32	.7z .cdt .dsk .sna .tap
Commodore 16	c16	RetroArch Vice	.cmd .crt `.cue and .bin` .d64 .d6z .d71 .d7z .d80 .d81 .d82 .d8z .g41 .g4z .g64 .g6z .m3u .nbz .nib .
Commodore 64	c64	RetroArch Vice	.cmd .crt `.cue and .bin` .d64 .d6z .d71 .d7z .d80 .d81 .d82 .d8z .g41 .g4z .g64 .g6z .m3u .nbz .nib .
Commodore VIC-20	vic20	RetroArch Vice	.cmd .crt `.cue and .bin` .d64 .d6z .d71 .d7z .d80 .d81 .d82 .d8z .g41 .g4z .g64 .g6z .m3u .nbz .nib .
DOS	dos	RetroArch DOSBox Pure	.7z .zip
NEC PC-98	pc98	RetroArch Neko Project II Kai	.2hd.88d.98d.cmd.d88.d98.dup.fdd.fdi .hdd .hdi .hdm .hdn .nhd .tfd .thd .xdi .zip
Sharp X68000	x68000	RetroArch PX68k	.2hd .88d .cmd .d8 .dim .dup .hdf .hdm .img .m3u .xdf.zip
ZX Spectrum	zxspectrum	RetroArch Fuse	.7z .rzx .scl .tap .trd .tzx .z80 .zip .zx

3.3.11 Sega Cheat Sheet

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System	ROM Folder Name	Emulator (Click for Link)	File Format	BIOS Files
Sega 32X	sega32x	RetroArch PicoDrive	.32x .7z .bin .zip	Not Required
Sega Atomiswave (Standalone)	atomiswave	**Flycast (Standalone)**	.zip	Place `awbios.zip` in `bios/flycast/bios`
Sega CD / Mega-CD	`segacd` or `megacd`	RetroArch Genesis Plus GX	.32x .7z .chd '.cue and .bin' .iso .m3u .zip	`bios_CD_E.bin` (Europe) `bios_CD_U.bin` (US) `bios_CD_J.bin` (Japan)
Sega Dreamcast	dreamcast	RetroArch FlyCast	.cdi .chd `.cue and .bin` `.gdi and .bin`	Create `dc` folder in `bios` Place `dc_boot.bin` in `bios/dc`
Sega Dreamcast (Standalone)	dreamcast	**Flycast (Standalone)**	.cdi .chd `.cue and .bin` `.gdi and .bin`	Not Required
Sega Game Gear	gamegear	RetroArch Genesis Plus GX	.7z .gg .zip	Not Required
Sega Genesis / Mega Drive	`genesis` or `megadrive`	RetroArch Genesis Plus GX	.7z .bin .gen .md .smd .zip	Not Required
Sega Genesis Widescreen	genesiswide	RetroArch Genesis Plus GX	.7z .bin .gen .md .smd .zip	Not Required
Sega Master System	mastersystem	RetroArch Genesis Plus GX	.7z .gen .sms .zip	Not Required
Sega Model 2	model2/roms	https://emulation.gametechwiki.com/index.php/Sega_Model_2	.zip	Not Required
Sega Model 3	model3	Supermodel	.zip	Not Required
Sega NAOMI	naomi	RetroArch FlyCast	.zip	Create `dc` folder in `bios` Place `naomi.zip` in `bios/dc`
Sega NAOMI (Standalone)	naomi	**Flycast (Standalone)**	.zip	Place `naomi.zip` in `bios/flycast/bios`
Sega NAOMI 2 (Standalone)	naomi2	**Flycast (Standalone)**	.zip	Place `naomi2.zip` in `bios/flycast/bios`
Sega Saturn	saturn	RetroArch Beetle Saturn	.7z .chd `.cue and .bin` .iso .zip	`sega_101.bin` (JP) `mpr-17933.bin` (US/EU)
Sega Saturn	saturn	RetroArch Kronos	.7z .chd `.cue and .bin` .iso .zip	Create `kronos` folder in `bios` Place `saturn_bios.bin` in `bios/kronos`
Sega Saturn	saturn	RetroArch Yabause	.7z .chd `.cue and .bin` .iso .zip	`saturn_bios.bin`

3.3.12 SNK Cheat Sheet

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System	ROM Folder Name	Emulator (Click for Link)	File Format	BIOS Files
SNK Neo Geo MVS	fbneo	RetroArch FinalBurn Neo	.7z .zip	`neocdz.zip` (US) `neogeo.zip` (JP)
SNK Neo Geo CD	`neogeocd` or `neogeocdjp`	RetroArch FinalBurn Neo	.chd	`neocdz.zip` (US) `neogeo.zip` (JP)
SNK Neo Geo CD (Standalone)	`neogeocd` or `neogeocdjp`	**MAME (Standalone)**	.chd	Place `neocdz.zip` in `roms/neogeocd` or `roms/neogeocdjp` (matching the location of your ROMs)
SNK Neo Geo Pocket	ngp	RetroArch Beetle NeoPop	.7z .ngc .ngp .zip	Not Required
SNK Neo Geo Pocket Color	ngpc	RetroArch Beetle NeoPop	.7z .ngc .ngp .zip	Not Required

- **SNK Neo Geo CD**
- **SNK Neo Geo Pocket**
- **SNK Neo Geo Pocket Color**

3.3.13 Sony Cheat Sheet

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System	ROM Folder Name	Emulator (Click for Link)	File Format	BIOS Files
Sony PlayStation (Standalone)	psx	**DuckStation (Standalone)**	.chd `.cue and .bin` .ecm .iso .m3u .mds .pbp	Example Set: `scph5500.bin` (JP) `scph5501.bin` (US) `scph5502.bin` (EU)
Sony PlayStation	psx	RetroArch Beetle HW	.chd `.cue and .bin` .ecm .iso .m3u .mds .pbp	Example Set: `scph5500.bin` (JP) `scph5501.bin` (US) `scph5502.bin` (EU)
Sony PlayStation	psx	RetroArch Swanstation	.chd `.cue and .bin` .ecm .iso .m3u .mds .pbp	Example Set: `scph5500.bin` (JP) `scph5501.bin` (US) `scph5502.bin` (EU)

Sony PlayStation 2 (Standalone)	ps2	**PCSX2 (Standalone)**	`.bin` .chd .cso .dump .gz .img .iso .mdf .nrg	Example Set (EU Set):
				`SCPH-70004_BIOS_V12_EUR_200`
				`SCPH-70004_BIOS_V12_EUR_200`
				`SCPH-70004_BIOS_V12_EUR_200`
				`SCPH-70004_BIOS_V12_EUR_200`
				
				Example Set (US Set):
				`SCPH-70012_BIOS_V12_USA_200`

Sony PlayStation 3 (Standalone)	ps3	**RPCS3 (Standalone)**	• Game Folder: `/PS3_GAME/USRDIR/eboot.bin` in `roms/ps3` • An installed `.pkg and .rap` file through RPCS3 • ISO file format is **not** supported	Firmware installation required through RPCS3 directly Read the Quickstart Guide for m information
Sony PlayStation Portable (Standalone)	psp	**PPSSPP (Standalone)**	.chd .cso .elf .iso .pbp .prx	Not Required
Sony PlayStation Portable	psp	RetroArch PPSSPP	.7z .chd .cso .elf .iso .pbp .prx	The RetroArch core requires `ppsspp.zip` in: `bios` `ppsspp.zip` is provided throug RetroArch's downloader. Open `RetroArch` Click `Online Updater` > `Core System Files Downloade` download `ppsspp.zip`
Sony PlayStation Vita (Standalone)	psvita	**Vita3K (Standalone)**	•Game Folder in `roms/psvita/ux0` •A valid installed ROM file through Vita3K (`.pkg and work.bin`, .zip)	Firmware installation required through Vita3K directly Read the Quickstart Guide for m information

May 27, 2026

November 16, 2025

4. Decompilations

4.1 Decompilations

4.1.1 Decompilation Table of Contents

4.1.2 Decompilations and Reverse Engineered PC Ports

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4.1.3 Getting Started with Decompilations and Reverse Engineered PC Ports

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How to Set up Distrobox

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Many of the decompilations and reverse engineered PC ports must be compiled. Compiling on the Steam Deck can be a bit tricky. Distrobox is one solution to make compiling these easy to do and any changes made within a Distrobox are retained on SteamOS updates.

This section will go over setting up a Distrobox, which you will utilize throughout this page.

PREREQUISITES: SUDO PASSWORD

Note: Skip this section if you have already set a sudo password

1. In Desktop Mode, open `Konsole`, type in `passwd`, and press enter
2. You'll be prompted to create a password. Your text will not be visible. After you press enter, you will need to type your password again to confirm
3. Exit out of Konsole

HOW TO INSTALL DISTROBOX

Note: This section will require using sudo.

1. In Desktop Mode, open Konsole and enter the following lines, one at a time

```
curl -s https://raw.githubusercontent.com/89luca89/distrobox/main/install | sh -s -- --prefix ~/.local
```

```
sudo touch /etc/subuid /etc/subgid
```

```
sudo usermod --add-subuid 100000-165535 --add-subgid 100000-165535 $USER
```

1. Download a supported container manager. For most people, lilipod is more than enough and can be obtained by downloading `lilipod-linux-amd64` from https://github.com/89luca89/lilipod/releases/latest
   - Alternatively, you can use podman if you need more than what lilipod offers
2. Rename `lilipod-linux-amd64` to `lilipod`. If using podman, rename `podman-launcher-amd64` to `podman`
   - Casing and spelling are important
3. Right click `lilipod` or `podman`, click `Properties`, click `Permissions`, and ensure `Is executable` is checked
4. Move the `lilipod` or `podman` file to `$HOME/.local/bin/`
   - `~/.local` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click "view hidden files" to see these folders

HOW TO CONFIGURE BASHRC

Note

These instructions are for the default shell
If you are using another shell, replace `.bashrc` with the rc of your appropriate shell

1. In Desktop Mode, open the `$HOME` folder
   - You may not see the word `deck` in the file path at the top, this is the `home` folder for the `deck` user
2. Click the hamburger menu in the top right, ☰, click `Show Hidden Files`
3. Right click `.bashrc`, click `Open with Kate` or a file editor of your choice
4. Add these lines to the bottom of the `.bashrc` file

```
# Uncomment the xhost line below if you know that you are using xhost
#xhost +si:localuser:$USER
export PATH=$HOME/.local/bin:$PATH
```

1. Save the `.bashrc` file and exit
2. If you had a terminal open previously (Konsole), close out of it and re-open before proceeding to the next section

HOW TO SET UP DISTROBOX

The various sections on this page will assume you are using the Ubuntu distrobox created in the section below. However, if you prefer, you may also create a Debian distrobox (the commands will mostly be identical) or use a distro of your choice (you will need to figure out how to install the various dependencies).

1. Open Konsole
2. Create a Distrobox of a distro of your choice. In this example, we will use Ubuntu:
   - `distrobox create --name ubuntu -i ubuntu:23.04`

   ```
   * `--name ubuntu` assigns a name to the distrobox, replace `ubuntu` with your preferred name
   * `-i ubuntu:23:04` selects a distro, replace `ubuntu:23:04` with your preferred distro
   * If you get multiple options after inputting this command, select the `docker.io/library/ubuntu:##.##` image by pressing enter on the respective
   line
   ```

3. To enter the Distrobox, open Konsole and enter:
   - `distrobox enter ubuntu`

   ```
   * This command enters the distrobox (assigned by name in Step 2), replace `ubuntu` with the name you typed in Step 2
   * You will need to enter the Distrobox when compiling the various games on this page. You can identify when you are in a Distrobox by looking at the
   lefthand side of Konsole. Using the Distrobox created by this guide, it will say `deck@ubuntu`
   ```

Other Distros

You can see a list of all available distros compatible with Distrobox here

---

Terminal Tips and Tricks

- `cd`
  - Change directories
  - For example, if you are working in a folder in terminal and need to change to a subfolder, enter `cd subfoldername`

  ```
  * You can press tab to auto-complete the sub-folder name as well
  ```

- `cd ..` or `cd -`
    - Change into the previous directory
- `Tab` button
    - Auto complete file and path names
- `ls` or `ll`
    - List all of the folders and files in the current directory
- `~` can be used in place of `home`
    - For example, if you have a folder in your `Applications` that you would like to navigate to, you can use `cd ~/Applications` instead of typing `cd $HOME/Applications`
- `mkdir`
    - Create folders
    - For example: `mkdir foldername`
- `Ctrl` + `C`
    - Terminate a running process
- `Up` arrow key button
    - Repeat last used command
- `Ctrl` + `L`
    - Clear the screen
- `Ctrl` + `U`
    - Clear current input
- `Ctrl` + `A`
    - Jump to beginning of the line

4.1.4 Games

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2Ship2Harkinian: Majora's Mask

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WHAT IS 2SHIP2HARKINIAN?

`An unofficial PC port of The Legend of Zelda Majora's Mask.`

Source: https://github.com/HarbourMasters/2ship2harkinian

SUPPORTED LEGEND OF ZELDA: MAJORA'S MASK ROMS

To check the hash of your ROM, right click the ROM, click `Properties`, click `Checksums`, click `Calculate` to the right of `SHA1` and validate it matches one of the below hashes.

- `7f5630dbc4d5d61d6276213210c4d5cdd83a47d6`
- `d6133ace5afaa0882cf214cf88daba39e266c078`
- `9743aa026e9269b339eb0e3044cd5830a440c1fd`

SETTING UP 2SHIP2HARKINIAN

Note: The following folder locations are recommendations. You can choose a different folder location.

1. In `$HOME/Applications`, create a `2Ship2Harkinian` folder
2. Download the latest `Linux` version of `2Ship2Harkinian`: https://github.com/HarbourMasters/2ship2harkinian/releases to the folder you created in Step 1
    - Download the `2Ship-VERSION-NAME-Linux.zip` file
3. Right click `2Ship-VERSION-NAME-Linux.zip` and click `Extract > Extract archive here`
    - If it creates a subfolder, move the contents directly to `$HOME/Applications/2Ship2Harkinian`
4. Right click `2ship.appimage`, click `Properties`, click `Permissions`, check `Is Executable`

INSTALLING 2SHIP2HARKINIAN

1. Place your `The Legend of Zelda: Majora's Mask` ROM in `$HOME/Applications/2Ship2Harkinian`
2. Open `2ship.AppImage`, wait for it to finish generating the OTR
3. To play 2Ship2Harkinian, open `2ship.AppImage`

HOW TO INSTALL CUSTOM TEXTURES AND MODS

Texture Pack and Mod Sources

This list is not exhaustive

- https://gamebanana.com
- https://evilgames.eu/texture-packs/oot-reloaded.htm

How to Install Custom Textures and Mods

1. In `$HOME/Applications/2Ship2Harkinian`, create a `mods` folder if one does not exist already
    - **Casing matters**
2. Place your mods or textures directly into this folder. Mods or textures typically have a `.otr` file extension
    - If you have a `.rar` or `.zip` file, you will need to extract it first
3. Depending on the mod or texture pack, you may need to enable it in game as well. Refer to the mod or texture pack for any additional instructions

2SHIP2HARKINIAN SAVE LOCATION

Saves for 2Ship2Harkinian are in the same folder as the `2ship.AppImage`. If you followed this section, your saves will be in `$HOME/Applications/2Ship2Harkinian`.

HOW TO TRANSFER ZELDA64: MAJORA'S MASK SAVES TO 2SHIP2HARKINIAN

If you have a save you would like to transfer from Zelda64: Majora's Mask, open 2Ship2Harkinian in Desktop Mode. Drag the save from Zelda64 to the open 2Ship2Harkinian window.

For the save locations, see Zelda64: Save Location

Cannonball: OutRun Engine

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HOW TO BUILD CANNONBALL ON THE STEAM DECK

WHAT IS THIS?

`CannonBall is a souped-up game engine for the OutRun arcade game.`

Source: https://github.com/djyt/cannonball

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
   - `distrobox enter ubuntu`
3. Enter the following:
   - `sudo apt install -y sdl2-dev libboost-dev cmake make`

SETTING UP CANNONBALL

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
   - `git clone --recursive https://github.com/djyt/cannonball.git`
3. A `cannonball` folder will be created, `
4. You will add your ROM files will be added after the build is finished

BUILDING CANNONBALL

1. In `$HOME/Applications/Distrobox/cannonball`, right click anywhere in the folder, click `Open Terminal Here`, enter:
   - `distrobox enter ubuntu`
2. Enter:
   - `mkdir build`
   - `mkdir roms`
   - `cd build`
   - `cmake --fresh -DTARGET=linux.cmake -DOpenGL_GL_PREFERENCE=GLVND -DCMAKE_BUILD_TYPE="Release" -S ../cmake/`
   - `make -j4 VERBOSE=1`
3. Wait for it to finish building

PROVIDING ROM FILES FOR SUPER CANNONBALL

You must provide particular versions of the OutRun ROM files to run Cannonball. OutRun Revision B Roms need to unpacked into the cannonball/build/roms folder. The cannonball/roms/roms.txt file lists the expected file names to help you check that you have unpacked the correct files.

The location of suitable files cannot be linked here because they are copyright. When you have a suitable archive unpack it so that the files are directly in cannonball/build/roms and not in any subdirectory.

HOW TO CONFIGURE CANNONBALL

See the Cannonball manual:
`https://github.com/djyt/cannonball/wiki/Cannonball-Manual`

No particular reconfiguration is needed for the Steam Deck. The game will default to 16:9 widescreen, and has no 16:10 mode, so there will be small black bars at the top and bottom of the SteamDeck screen. Options can be changed inside the Cannonball UI.

HOW TO RUN CANNONBALL

- `cd ~/Applications/distrobox/cannonball/build`
- `./cannonball`

OpenGOAL: Jak and Dexter: The Precursor Legacy

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WHAT IS OPENGOAL: JAK AND DEXTER: THE PRECURSOR LEGACY?

`OpenGOAL is an unofficial port of Jak and Daxter: The Precursor Legacy for Windows and Linux.`

Source: https://opengoal.dev/

SETTING UP OPENGOAL

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create an `OpenGOAL` folder
2. Place your `Jak and Dexter: The Precursor Legacy` ISO directly in the newly created `OpenGOAL` folder
3. Download the latest `amd64.AppImage` from here: https://github.com/open-goal/launcher/releases/latest and place it in the folder you created in Step 1
    - Example: Download `open-goal-launcher_#.#.#_amd64.AppImage`

    ```
    * The version number may be different depending on when you are reading this
    ```

4. Right click `open-goal-launcher_#.#.#_amd64.AppImage`, click `Properties`, click `Permissions`, check `Is Executable`

INSTALLING OPENGOAL

1. In `$HOME/Applications/OpenGOAL`, double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it
2. Click `Set Version`, click the `Download` icon on the row with the latest version at the top
3. Click the circle icon on the left to select the version and click `Save` in the top right
4. On the left side, select `Jak and Dexter`
5. On the bottom right, click `Install via ISO` and select your `Jak and Dexter: The Precursor Legacy` ISO
6. The application will begin to decompile your game, this may take a few moments
7. Once it is finished decompiling, press `Continue`
8. To play your game, open this launcher, select `Jak and Dexter` on the left, and click `Play` in the bottom right
    - The launcher does not work in Game Mode. See How to Add OpenGOAL: Jak and Dexter: The Precursor Legacy to Steam to learn how to play the game directly in Steam

HOW TO INSTALL CUSTOM TEXTURES

Note: Texture packs need to be zipped in order to be installed from the launcher. This section will use `texture_replacements.zip` from https://www.youtube.com/watch?v=lX1gBO1INZ4 as an example.

1. Download the custom textures from the pinned comment on this Youtube Video: https://www.youtube.com/watch?v=lX1gBO1INZ4 to `$HOME/Applications/OpenGOAL`
2. Double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it
3. On the left side, select `Jak and Dexter`
4. On the bottom right, select `Features`, `Texture Packs`
5. Click `Add New Pack`
6. Select `texture_replacements.zip` and wait a few moments
7. Your texture pack will appear below in its own box. Click the red `Disabled` button to enable your texture pack
8. Click `Apply Texture Changes` in the top right and wait a few moments
9. Your custom textures will now be installed

HOW TO ADD OPENGOAL: JAK AND DEXTER: THE PRECURSOR LEGACY TO STEAM

1. In `$HOME/Applications/OpenGOAL`, right click anywhere in the blank space, click `Create New` > `Text File`
2. Name it `Jak and Dexter: The Precursor Legacy.sh`
3. At the top of the text file, write `#!/bin/bash`
4. In `$HOME/Applications/OpenGOAL`, double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it

5. On the left side, select `Jak and Dexter`
6. On the bottom right, click the `Gear` icon, and click `Copy Game Executable Command`
7. In the text file you created in Step 1, right click anywhere below `#!/bin/bash` and click `Paste`
8. Save the text file and exit out
9. Right click your newly created text file, click `Properties`, click `Permissions`, check `Is Executable`
10. Right click the text file, click `Add to Steam`
    - If you are using EmulationStation-DE or Steam ROM Manager, place in `Emulation/roms/desktop` instead

```
* See [How to Add Decompilations and Reverse Engineered PC Ports to Steam](#how-to-add-decompilations-and-reverse-engineered-pc-ports-to-steam) for
further detail
```

Your text file should look similar to the text below:

Text file name:

`Jak and Dexter: The Precursor Legacy.sh`

Text file contents:

```
#!/bin/bash

$HOME/Applications/OpenGOAL/versions/official/v0.2.0/gk -v --proj-path $HOME/Applications/OpenGOAL/active/jak1/data --game jak1 -- -boot -fakeiso
```

Note: You will need to update this shortcut whenever you update OpenGOAL.

OpenGOAL: Jak II

PREFACE

The folders in this section will be repeated from OpenGOAL: Jak and Dexter: The Precursor Legacy. If you have already installed Jak and Dexter: The Precursor Legacy, you may use the same folders and launcher to install Jak II.

WHAT IS OPENGOAL: JAK II?

`OpenGOAL is an unofficial port of Jak II for Windows and Linux.`

Source: https://opengoal.dev/

SETTING UP OPENGOAL

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create an `OpenGOAL` folder
2. Place your `Jak II` ISO directly in the newly created `OpenGOAL` folder
3. Download the latest `amd64.AppImage` from here: https://github.com/open-goal/launcher/releases/latest and place it in the folder you created in Step 1
    - Example: Download `open-goal-launcher_#.#.#_amd64.AppImage`

```
* The version number may be different depending on when you are reading this
```

4. Right click `open-goal-launcher_#.#.#_amd64.AppImage`, click `Properties`, click `Permissions`, check `Is Executable`

INSTALLING OPENGOAL

If you already installed OpenGOAL: Jak and Dexter: The Precursor Legacy, you do not need to do Steps 2 and 3 again.

1. In `$HOME/Applications/OpenGOAL`, double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it
2. Click `Set Version`, click the `Download` icon on the row with the latest version at the top
3. Click the circle icon on the left to select the version and click `Save` in the top right
4. On the left side, select `Jak II`
5. On the bottom right, click `Install via ISO` and select your `Jak II` ISO
6. The application will begin to decompile your game, this may take a few moments
7. Once it is finished decompiling, press `Continue`
8. To play your game, open this launcher, select `Jak II` on the left, and click `Play` in the bottom right
    - The launcher does not work in Game Mode. See How to Add OpenGOAL: Jak II to Steam to learn how to play the game directly in Steam

HOW TO INSTALL CUSTOM TEXTURES

Note: Texture packs need to be zipped in order to be installed from the launcher.

1. Download a texture pack of your choice
2. Double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it
3. On the left side, select `Jak II`
4. On the bottom right, select `Features`, `Texture Packs`
5. Click `Add New Pack`
6. Select your zipped texture pack and wait a few moments
7. Your texture pack will appear below in its own box. Click the red `Disabled` button to enable your texture pack
8. Click `Apply Texture Changes` in the top right and wait a few moments
9. Your custom textures will now be installed

HOW TO ADD OPENGOAL: JAK II TO STEAM

1. In `$HOME/Applications/OpenGOAL`, right click anywhere in the blank space, click `Create New` > `Text File`
2. Name it `Jak II.sh`
3. At the top of the text file, write `#!/bin/bash`
4. In `$HOME/Applications/OpenGOAL`, double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it
5. On the left side, select `Jak II`
6. On the bottom right, click the `Gear` icon, and click `Copy Game Executable Command`
7. In the text file you created in Step 1, right click anywhere below `#!/bin/bash` and click `Paste`
8. Save the text file and exit out
9. Right click your newly created text file, click `Properties`, click `Permissions`, check `Is Executable`
10. Right click the text file, click `Add to Steam`
    - If you are using EmulationStation-DE or Steam ROM Manager, place in `Emulation/roms/desktop` instead

    ```
    * See [How to Add Decompilations and Reverse Engineered PC Ports to Steam](#how-to-add-decompilations-and-reverse-engineered-pc-ports-to-steam) for further detail
    ```

Your text file should look similar to the text below:

Text file name:

`Jak II.sh`

Text file contents:

```
#!/bin/bash
$HOME/Applications/OpenGOAL/versions/official/v0.2.0/gk -v --proj-path $HOME/Applications/OpenGOAL/active/jak2/data --game jak2 -- -boot -fakeiso
```

Note: You will need to update this shortcut whenever you update OpenGOAL.

Perfect Dark

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WHAT IS THIS DECOMPILATION?

`A work-in-progress port of the Perfect Dark decompilation to modern platforms.`

Source: https://github.com/fgsfdsfgs/perfect_dark

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following, one line at a time:
    - `sudo dpkg --add-architecture i386`
    - `sudo apt-get update`
4. Enter the following command:
    - `sudo apt-get install git gcc g++ gcc-multilib g++-multilib make libsdl2-dev zlib1g-dev libsdl2-dev:i386 zlib1g-dev:i386`

SETTING UP PERFECT DARK

Note: The following folder locations are recommendations. You may choose a different folder location.

1. Download the `i686-linux` version from https://github.com/fgsfdsfgs/perfect_dark#download to your `Downloads` folder
2. Right click `pd-i686-linux.zip`, click `Extract archive here, autodetect subfolder`
3. Move the newly extracted `pd-i686-linux` folder to `$HOME/Applications`
4. Right click `pd`
5. Click `Properties`
6. Click `Permissions`
7. Check `Is Executable`
8. Place your `Perfect Dark` ROM in `$HOME/Applications/pd-i686-linux/data`
    - MD5 Hash: `e03b088b6ac9e0080440efed07c1e40f`
    - To locate your MD5 Hash, right click your ROM, click `Properties`, click `Checksums`, click `Calculate` to the left of `MD5` and compare it to the above hash. If it is a match, you have a valid ROM 5
9. In order to properly launch Perfect Dark, proceed to the next section

HOW TO LAUNCH PERFECT DARK

1. Download attached `.sh` file
    - PerfectDark.sh
    - **Note:** If you are using different folder locations, make sure to edit the above file and edit the paths

2. Place in `$HOME/Applications`
    - If you are using EmulationStation-DE or Steam ROM Manager, place in `Emulation/roms/desktop` instead

    ```
    * See [How to Add Decompilations and Reverse Engineered PC Ports to Steam](#how-to-add-decompilations-and-reverse-engineered-pc-ports-to-steam) for instructions
    ```

3. Right click `PerfectDark.sh`
4. Click `Properties`
5. Click `Permissions`
6. Check `Is Executable`
7. Use `PerfectDark.sh` to open Perfect Dark

HOW TO CONFIGURE PERFECT DARK

1. Open `PerfectDark.sh` at least once so it can generate the `pd.ini` file
2. Open the `$HOME/.local/share/perfectdark/` folder
    - `~/.local` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click `Show Hidden Files` to see these folders
3. Right click `pd.ini`, click `Open with Kate` or a text editor of your choice
4. Customize settings

Recommended Settings

- Set `DefaultWidth` to `800`
- Set `DefaultHeight` to `1200`
- Set `DefaultFullscreen` to `1`

CONTROLS

https://github.com/fgsfdsfgs/perfect_dark#controls

Render96ex

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HOW TO COMPILE RENDER96EX ON THE STEAM DECK

WHAT IS RENDER96EX?

`An "HD" version of Super Mario 64 based on the rendered advertisements and art published in Nintendo Power.`

Source: https://github.com/Render96/Render96ex/wiki

When you include the HD textures as part of this guide the whole Render96ex install will need 1.1GB of storage. This compares to less than 0.1 GB for the SM64Plus version of the game.

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following:
    - `sudo apt install -y git build-essential pkg-config libusb-1.0-0-dev libsdl2-dev bsdmainutils libglew-dev`

SETTING UP RENDER96EX

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone --single-branch --branch alpha https://github.com/Render96/Render96ex.git`
3. A `Render96ex` folder will be created, place your Super Mario 64 ROM in this folder
4. Rename the Super Mario 64 ROM to `baserom.us.z64`

BUILDING RENDER96EX

1. In `$HOME/Applications/Distrobox/Render96ex/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make -j4 TEXTURE_FIX=1`
3. Wait for it to finish building
4. To play Render96ex, open `sm64.us.f3dex2e` in `$HOME/Applications/Distrobox/Render96ex/build/us_pc`

HOW TO INSTALL CUSTOM MODELS

1. Download the latest model pack: https://github.com/Render96/ModelPack/releases to `$HOME/Downloads`
2. Right click `Render96_DynOs_v3.2.7z` and click `Extract archive here, autodetect subfolder`
3. Move the newly extracted `Render96_DynOs...` folder to `$HOME/Applications/Distrobox/Render96ex/build/us_pc/dynos/packs`
4. To enable custom models, in game, press `Start`, press `L2`, select `Model Packs` and enable `Render96DynOs...`

HOW TO INSTALL CUSTOM TEXTURES

1. In `$HOME/Applications/Distrobox/Render96ex/build/us_pc/res`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone https://github.com/pokeheadroom/RENDER96-HD-TEXTURE-PACK.git -b master`
2. The `RENDER96-HD-TEXTURE-PACK` is now installed

HOW TO CONFIGURE RENDER96EX

GUI

1. While in game, press `Start`, press `R1`
2. Configure settings

Config File

1. Open `sm64.us.f3dex2e` in `$HOME/Applications/Distrobox/Render96ex/build/us_pc` at least once so it can generate the `sm64config.txt` file
2. Open the `$HOME/.local/share/sm64ex/` folder
    - `~/.local` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click `Show Hidden Files` to see these folders
3. Right click `sm64config.txt`, click `Open with Kate` or a text editor of your choice
4. Customize settings

Ship of Harkinian: Ocarina of Time

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WHAT IS SHIP OF HARKINIAN?

`An unofficial PC port of The Legend of Zelda Ocarina of Time.`

Source: https://www.shipofharkinian.com/

SUPPORTED LEGEND OF ZELDA: OCARINA OF TIME ROMS

For a full list of supported ROMs and a ROM validation tool, see https://ship.equipment/.

To confirm that you have a valid ROM, drag and drop your ROM to the website above.

SETTING UP SHIP OF HARKINIAN

Note: The following folder locations are recommendations. You can choose a different folder location.

1. In `$HOME/Applications`, create a `ShipofHarkinian` folder
2. Download the latest `Linux-Performance` version of `Ship of Harkinian`: https://github.com/HarbourMasters/Shipwright/releases to the folder you created in Step 1
    - Download the `SoH-VERSION-NAME-Linux-Performance.zip` file
3. Right click the downloaded zip file and click `Extract archive here, detect subfolder`
4. Move the `soh.AppImage` to `$HOME/Applications/ShipofHarkinian`
5. Right click `soh.AppImage`, click `Properties`, click `Permissions`, check `Is Executable`

INSTALLING SHIP OF HARKINIAN

1. Place your `The Legend of Zelda: Ocarina of Time` ROM in `$HOME/Applications/ShipofHarkinian`
2. Open `soh.AppImage`, wait for it to finish generating the OTR
3. To play Ship of Harkinian, open `soh.AppImage`

HOW TO INSTALL CUSTOM TEXTURES AND MODS

Texture Pack and Mod Sources

This list is not exhaustive

- https://gamebanana.com/games/16121
- https://evilgames.eu/texture-packs/oot-reloaded.htm

How to Install Custom Textures and Mods

1. In your Ship of Harkinian folder, create a `mods` folder, all lowercase
    - `$HOME/Applications/ShipofHarkinian` if you used the folders in the above guide
    - **Casing matters**
2. Place your mods or textures directly into this folder. Mods or textures typically have a `.otr` file extension
    - If you have a `.rar` or `.zip` file, you will need to extract it first
3. Depending on the mod or texture pack, you may need to enable it in game as well. Refer to the mod or texture pack for any additional instructions

SHIP OF HARKINIAN SAVE LOCATION

Saves for Ship of Harkinian are in the same folder as the `soh.AppImage`. If you followed this section, your saves will be in `$HOME/Applications/ShipofHarkinian`.

---

sm64ex

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WHAT IS THIS DECOMPILATION?

`A fork of sm64-port/sm64-port with additional features.`

Source: https://github.com/sm64pc/sm64ex

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following:
    - `sudo apt install build-essential git python3 libglew-dev libsdl2-dev bsdmainutils patch`

SETTING UP SM64EX

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone https://github.com/sm64pc/sm64ex`
3. A `sm64ex` folder will be created, place your Super Mario 64 ROM in this folder
4. Rename the Super Mario 64 ROM to `baserom.us.z64`

BUILDING SM64EX

1. In the `sm64ex` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make`
3. Wait for it to finish building
4. To play `sm64ex`, open `sm64.us.f3dex2e` in `$HOME/Applications/Distrobox/sm64ex/build/us_pc`

HOW TO CONFIGURE SM64EX

1. Open `sm64.us.f3dex2e` in `$HOME/Applications/Distrobox/sm64ex/build/us_pc` at least once so it can generate the `sm64config.txt` file
2. Open the `$HOME/.local/share/sm64ex/` folder
    - `~/.local` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click `Show Hidden Files` to see these folders
3. Right click `sm64config.txt`, click `Open with Kate` or a text editor of your choice
4. Customize settings

HOW TO APPLY PATCHES

A 60 FPS patch is included with the GitHub repo for sm64ex. However, you can take what's written here and apply additional patches to sm64ex.

1. In the `$HOME/Applications/Distrobox/sm64ex/enhancements` folder, right click anywhere in the folder, click `Open Terminal Here`
2. Enter:
    - `git apply 60fps_ex.patch --ignore-whitespace --reject`

    ```
    * `60fps_ex.patch` is the name of the included file. If the file name is different, replace the file name in the above command
    ```

3. You will need to re-build sm64ex for the patches to apply. To do so, enter the Ubuntu Distrobox and run `make clean` in the `` `$HOME/Applications/Distrobox/sm64ex `` folder

---

sm64ex-coop

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WHAT IS THIS DECOMPILATION?

`Online multiplayer mod for SM64 that synchronizes all entities and every level for multiple players. Fork of sm64pc/sm64ex.`

Source: https://github.com/djoslin0/sm64ex-coop

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following:
    - `sudo apt install build-essential git python3 libglew-dev libsdl2-dev libz-dev bsdmainutils`

SETTING UP SM64EX-COOP

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone https://github.com/djoslin0/sm64ex-coop.git`
3. A `sm64ex-coop` folder will be created, place your Super Mario 64 ROM in this folder
4. Rename the Super Mario 64 ROM to `baserom.us.z64`

BUILDING SM64EX-COOP

1. In the `sm64ex-coop` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make -j4`
3. Wait for it to finish building
4. To play `sm64ex-coop`, open `sm64.us.f3dex2e` in `$HOME/Applications/Distrobox/sm64ex-coop/build/us_pc`

HOW TO CONFIGURE SM64EX-COOP

1. Open `sm64.us.f3dex2e` in `$HOME/Applications/Distrobox/sm64ex-coop/build/us_pc`
2. On the main menu, click `Options`

HOW TO PLAY MULTIPLAYER

See https://github.com/djoslin0/sm64ex-coop/wiki/Hosting-and-Joining.

Sonic 1 and 2

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PREFACE

To compile Sonic 1 and 2, you will need a legally obtained copy of a `Data.rsdk` file, obtained either from the Sonic 1 and 2 versions on the Android or iOS versions. For instructions, see Android and iOS.

WHAT IS THIS DECOMPILATION?

`A complete decompilation of Retro Engine v4 and the menus from Sonic 1 and 2 (2013).`

Source: https://github.com/Rubberduckycooly/Sonic-1-2-2013-Decompilation

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`

3. Enter the following:
    - `sudo apt install build-essential git libsdl2-dev libvorbis-dev libogg-dev libglew-dev libdecor-0-dev`

SETTING UP SONIC 1 AND 2

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone --recursive https://github.com/Rubberduckycooly/Sonic-1-2-2013-Decompilation.git`
3. A `Sonic-1-2-2013-Decompilation` folder will be created

BUILDING SONIC 1 AND 2

1. In the `Sonic-1-2-2013-Decompilation` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make -j5`
3. In the `Sonic-1-2-2013-Decompilation`, a `bin` folder will be created
4. If you would like to play both Sonic 1 and 2, make a copy of the `bin` folder directly in the `Sonic-1-2-2013-Decompilation` folder
5. Rename the original `bin` and the copied folder to Sonic 1 and Sonic 2 respectively
    - This step is for easier folder management, it is a recommendation and not a requirement
6. Place the matching `Data.rsdk` in each folder
7. To play Sonic 1 and 2, open `RSDKv4`

HOW TO CONFIGURE SONIC 1 AND 2

1. In the respective Sonic 1 and 2 folder, after launching `RSDKv4` for the first time, a `settings.ini` will be created
2. Right click `settings.ini`, click `Open with Kate` or a text editor of your choice
3. Customize settings

HOW TO MOD SONIC 1 AND 2

Mod Resources

This list is not comprehensive

- Sonic 1: GameBanana
- Sonic 2: GameBanana

1. In the respective `Sonic 1` or `Sonic 2` folders, create a `mod` folder
2. Download and extract mod(s) to the `mod` folder
    - A mod folder should typically have a `Data` folder and a `mod.ini` file
3. Right click `settings.ini`, click `Open with Kate` or a text editor of your choice
4. Change `DevMenu=false` to `DevMenu=true` and save and close out of the file
5. While in game, press `Start` and you should see a list of your mod(s)
6. To activate/toggle a mod, press `A` on the respective mod

---

Sonic CD

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PREFACE

To compile Sonic CD, you will need a legally obtained copy of a `Data.rsdk` file, obtained either from the Sonic CD versions on the Android or iOS versions. For instructions, see Android and iOS.

WHAT IS THIS DECOMPILATION?

`A complete decompilation of Retro Engine v3/Sonic CD.`

Source: https://github.com/Rubberduckycooly/Sonic-CD-11-Decompilation

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following command:
    - `sudo apt install build-essential git libsdl2-dev libvorbis-dev libogg-dev libtheora-dev libglew-dev`

SETTING UP SONIC CD

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone --recursive https://github.com/Rubberduckycooly/Sonic-CD-11-Decompilation.git`
3. A `Sonic-CD-11-Decompilation` folder will be created

BUILDING SONIC CD

1. In the `Sonic-CD-11-Decompilation` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make CXXFLAGS=-O2 -j5`
3. In the `Sonic-CD-11-Decompilation`, a `bin` folder will be created
4. Place the `Data.rsdk` you retrieved from How to Compile Sonic CD on the Steam Deck directly into the `bin` folder
5. To play Sonic CD, open `RSDKv4`

HOW TO CONFIGURE SONIC CD

1. In the `bin`, after launching `RSDKv4` for the first time, a `settings.ini` will be created
2. Right click `settings.ini`, click `Open with Kate` or a text editor of your choice
3. Customize settings

HOW TO MOD SONIC CD

Mod Resources

This list is not comprehensive

- https://gamebanana.com/mods/games/6108

1. In the `bin` folders, create a `mod` folder
2. Download and extract mod(s) to the `mod` folder
    - A mod folder should typically have a `Data` folder and a `mod.ini` file
3. Right click `settings.ini`, click `Open with Kate` or a text editor of your choice
4. Change `DevMenu=false` to `DevMenu=true` and save and close out of the file
5. While in game, press `Start` and you should see a list of your mod(s)
6. To activate/toggle a mod, press `A` on the respective mod

Super Mario 64 Plus

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HOW TO COMPILE SUPER MARIO 64 PLUS ON THE STEAM DECK

WHAT IS SUPER MARIO 64 PLUS?

```
SM64Plus is a fork of sm64-port that focuses on customizability and aims to add features that not only fix some of the issues
found in the base game but also enhance the gameplay overall with extra options.
```

Source: https://github.com/MorsGames/sm64plus

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following command:
    - `sudo apt install -y git build-essential pkg-config libusb-1.0-0-dev libsdl2-dev bsdmainutils`

SETTING UP SUPER MARIO 64 PLUS

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone https://github.com/MorsGames/sm64plus`
3. A `sm64plus` folder will be created, place your Super Mario 64 ROM in this folder
4. Rename the Super Mario 64 ROM to `baserom.us.z64`

BUILDING SUPER MARIO 64 PLUS

1. In `$HOME/Applications/Distrobox/sm64plus/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make -j4`
3. Wait for it to finish building
4. To play Super Mario 64 Plus, open `sm64.us` in `$HOME/Applications/Distrobox/sm64plus/build/us_pc`
    - The game may have missing HUD UI textures, to fix these, read the next section

HOW TO FIX THE MISSING HUD UI TEXTURES

1. Download attached `.sh` file
    - SuperMario64Plus.sh
    - **Note:** If you are using different folder locations, make sure to edit the above file and edit the paths
2. Place in `$HOME/Applications`
    - If you are using EmulationStation-DE or Steam ROM Manager, place in `Emulation/roms/desktop` instead

    ```
    * See [How to Add Decompilations and Reverse Engineered PC Ports to Steam](#how-to-add-decompilations-and-reverse-engineered-pc-ports-to-steam) for
    instructions
    ```

3. Right click `SuperMario64Plus.sh`
4. Click `Properties`
5. Click `Permissions`

6. Check `Is Executable`
7. Use `SuperMario64Plus.sh` to open SM64Plus

HOW TO CONFIGURE SUPER MARIO 64 PLUS

1. Open `sm64.us` in `$HOME/Applications/Distrobox/sm64plus/build/us_pc` at least once so it can generate the `settings.ini` file
2. Open the `$HOME/Applications/Distrobox/sm64plus/build/us_pc` folder
3. Right click `settings.ini`, click `Open with Kate` or a text editor of your choice
4. Customize settings

Recommended Settings

- Set `window_height` to `800`

HOW TO ADD SUPER MARIO 64 PLUS TO STEAM

1. In `$HOME/Applications`, right click `SuperMario64Plus.sh`, click `Add to Steam`
2. After adding it to Steam, you may rename the shortcut in Steam directly

---

Super Mario Bros

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HOW TO COMPILE SUPER MARIO BROS ON THE STEAM DECK

Note: This section requires a legal copy of `Super Mario World` and `Super Mario All-Stars` ROMs for the SNES.

WHAT IS THIS?

`A reverse engineered clone of Super Mario Bros.`

Source: https://github.com/snesrev/smw

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following command:
    - `sudo apt install libsdl2-dev python3-pip make python3-zstandard`

HOW TO SET UP SUPER MARIO BROS

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
   - `git clone https://github.com/snesrev/smw smb1`
3. A `smb1` folder will be created, place your `Super Mario All-Stars` ROM in `$HOME/Applications/smb1/other`
   - SHA256 Hash: `c05817c5b7df2fbfe631563e0b37237156a8f6b6`
   - To locate your SHA256 Hash, right click your ROM, click `Properties`, click `Checksums`, click `Calculate` to the left of `SHA1` and compare it to the above hash. If it is a match, you have a valid ROM
4. Rename the Super Mario World ROM to `smas.sfc`
5. Place your `Super Mario World` ROM in `$HOME/Distrobox/Applications/smbll`
   - SHA256 Hash: `6b47bb75d16514b6a476aa0c73a683a2a4c18765`
   - To locate your SHA256 Hash, right click your ROM, click `Properties`, click `Checksums`, click `Calculate` to the left of `SHA1` and compare it to the above hash. If it is a match, you have a valid ROM
6. Rename the `Super Mario World` ROM to `smw.sfc`

HOW TO BUILD SUPER MARIO BROS

1. In `$HOME/Applications/Distrobox/smb1/other`, right click anywhere in the folder, click `Open Terminal Here`, enter:
   - `distrobox enter ubuntu`
2. Enter:
   - `python3 extract.py`
3. Change directories to `$HOME/Applications/Distrobox/smb1` by entering the following:
   - `cd ..`
   - If you are not comfortable with the terminal, you may also open a terminal in `$HOME/Applications/Distrobox/smb1`, enter the Distrobox again by entering: `distrobox enter ubuntu`
4. Enter:
   - `make`
5. Wait for it to finish building

HOW TO PLAY SUPER MARIO BROS

1. Download attached `.sh` file
   - smb1.sh
   - **Note:** If you are using different folder locations, make sure to edit the above file and edit the paths
2. Place in `$HOME/Applications`
3. Right click `smb1.sh`
4. Click `Properties`
5. Click `Permissions`
6. Check `Is Executable`
7. Use `smb1.sh` to open Super Mario Bros

HOW TO CONFIGURE SUPER MARIO BROS

1. In `$HOME/Applications/Distrobox/smb1/`, right click `smw.ini`, click `Open with Kate` or a text editor of your choice
2. Customize settings

Recommended

- Set `Fullscreen` to `1`

Super Mario Bros: The Lost Levels

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HOW TO COMPILE SUPER MARIO BROS: THE LOST LEVELS ON THE STEAM DECK

Note: This section requires a legal copy of `Super Mario World` and `Super Mario All-Stars` ROMs for the SNES.

WHAT IS THIS?

`A reverse engineered clone of Super Mario Bros: The Lost Levels.`

Source: https://github.com/snesrev/smw

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
   - `distrobox enter ubuntu`
3. Enter the following command:
   - `sudo apt install libsdl2-dev python3-pip make git python3-zstandard`

HOW TO SET UP SUPER MARIO BROS: THE LOST LEVELS

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
   - `git clone https://github.com/snesrev/smw smbll`
3. A `smbll` folder will be created, place your `Super Mario All-Stars` ROM in `$HOME/Distrobox/Applications/smbll/other`
   - SHA256 Hash: `c05817c5b7df2fbfe631563e0b37237156a8f6b6`
   - To locate your SHA256 Hash, right click your ROM, click `Properties`, click `Checksums`, click `Calculate` to the left of `SHA1` and compare it to the above hash. If it is a match, you have a valid ROM
4. Rename the `Super Mario All-Stars` ROM to `smas.sfc`
5. Place your `Super Mario World` ROM in `$HOME/Distrobox/Applications/smbll`
   - SHA256 Hash: `6b47bb75d16514b6a476aa0c73a683a2a4c18765`
   - To locate your SHA256 Hash, right click your ROM, click `Properties`, click `Checksums`, click `Calculate` to the left of `SHA1` and compare it to the above hash. If it is a match, you have a valid ROM
6. Rename the `Super Mario World` ROM to `smw.sfc`

HOW TO BUILD SUPER MARIO BROS: THE LOST LEVELS

1. In `$HOME/Applications/Distrobox/smbll/other`, right click anywhere in the folder, click `Open Terminal Here`, enter:
   - `distrobox enter ubuntu`
2. Enter:
   - `python3 extract.py`
3. Change directories to `$HOME/Applications/Distrobox/smbll` by entering the following:
   - `cd ..`
   - If you are not comfortable with the terminal, you may also open a terminal in `$HOME/Applications/Distrobox/smbll`, enter the Distrobox again by entering: `distrobox enter ubuntu`
4. Enter:
   - `make`

5. Wait for it to finish building

HOW TO PLAY SUPER MARIO BROS: THE LOST LEVELS

1. Download attached `.sh` file
    - smbll.sh
    - **Note:** If you are using different folder locations, make sure to edit the above file and edit the paths
2. Place in `$HOME/Applications`
3. Right click `smbll.sh`
4. Click `Properties`
5. Click `Permissions`
6. Check `Is Executable`
7. Use `smbll.sh` to open Super Mario Bros: The Lost Levels

HOW TO CONFIGURE SUPER MARIO BROS: THE LOST LEVELS

1. In `$HOME/Applications/Distrobox/smbll/`, right click `smw.ini`, click `Open with Kate` or a text editor of your choice
2. Customize settings

Recommended

- Set `Fullscreen` to `1`

Super Mario World

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HOW TO COMPILE SUPER MARIO WORLD ON THE STEAM DECK

WHAT IS THIS?

`A reverse engineered clone of Super Mario World.`

Source: https://github.com/snesrev/smw

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following command:
    - `sudo apt install libsdl2-dev python3-pip make git python3-zstandard`

HOW TO SET UP SUPER MARIO WORLD

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone https://github.com/snesrev/smw`
3. A `smw` folder will be created, place your Super Mario World ROM in `$HOME/Applications/smw`
    - SHA256 Hash: `6b47bb75d16514b6a476aa0c73a683a2a4c18765`
    - To locate your SHA256 Hash, right click your ROM, click `Properties`, click `Checksums`, click `Calculate` to the left of `SHA1` and compare it to the above hash. If it is a match, you have a valid ROM
4. Rename the Super Mario World ROM to `smw.sfc`

HOW TO BUILD SUPER MARIO WORLD

1. In `$HOME/Applications/Distrobox/smw/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make`
3. Wait for it to finish building
4. To play Super Mario World, open `smw` in `$HOME/Applications/Distrobox/smw/`

HOW TO CONFIGURE SUPER MARIO WORLD

1. In `$HOME/Applications/Distrobox/smw/`, right click `smw.ini`, click `Open with Kate` or a text editor of your choice
2. Customize settings

Recommended

- Set `Fullscreen` to `1`

HOW TO MOD SUPER MARIO WORLD

1. In `$HOME/Applications/Distrobox/smw/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `git clone https://github.com/snesrev/smw_hacks.git`
3. Enter:
    - `pip install --break-system-packages --user bsdiff4`

4. Select a mod:
    - You do not need to download any additional content from the links below. They are strictly for additional context on what the mod entails
    - You can only have one mod active at a time. These mods will override the base Super Mario World game
    - Return to Dinosaur Land

```
* Enter:
    * `python3 assets/restool.py --hack=return_dino_land`
```

    - SMW with Levels from NSMB

```
* Enter:
    * `python3 assets/restool.py --hack=nsmb`
```

    - Super Mario World Enhanced

```
* Enter:
    * `python3 assets/restool.py --hack=enhanced`
```

    - Super Mario World Redrawn

```
* Enter:
    * `python3 assets/restool.py --hack=redrawn`
```

    - To remove a mod and play vanilla Super Mario World

```
* Enter:
    * `python3 assets/restool.py`
```

5. To play the mod you selected, open `smw` in `$HOME/Applications/Distrobox/smw/`

Super Metroid

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HOW TO BUILD SUPER METROID ON THE STEAM DECK

WHAT IS THIS?

`A reverse engineered clone of Super Metroid.`

Source: https://github.com/snesrev/sm

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following:
    - `sudo apt install libsdl2-dev python3-pip make git`

SETTING UP SUPER METROID

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone --recursive https://github.com/snesrev/sm`
3. A `sm` folder will be created, place your US Super Metroid ROM in `$HOME/Applications/sm`
4. Rename the Super Metroid ROM to `sm.smc`

BUILDING SUPER METROID

1. In `$HOME/Applications/Distrobox/sm`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make`
3. Wait for it to finish building
4. To play Super Metroid, open `sm` in `$HOME/Applications/Distrobox/sm/`

HOW TO CONFIGURE SUPER METROID

1. In `$HOME/Applications/Distrobox/sm/`, right click `sm.ini`, click `Open with Kate` or a text editor of your choice
2. Customize settings

Super Metroid Redux

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HOW TO BUILD SUPER METROID REDUX ON THE STEAM DECK

WHAT IS THIS?

A port of `Super Metroid Redux`: https://www.romhacking.net/hacks/4963/, using `a reverse engineered clone of Super Metroid.`

Source: https://github.com/testyourmine/sm-redux

Source: https://github.com/snesrev/sm

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following:
    - `sudo apt install libsdl2-dev python3-pip make git`

PATCHING SUPER METROID REDUX

1. Download the patch here: https://www.romhacking.net/hacks/4963/
2. Use the romhacking website here to patch your ROM: https://www.romhacking.net/patch/

SETTING UP SUPER METROID REDUX

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone --recursive https://github.com/testyourmine/sm-redux`
3. A `sm-redux` folder will be created, place your patched US Super Metroid ROM in `$HOME/Applications/sm-redux`
4. Rename the patched Super Metroid ROM to `sm.smc`

BUILDING SUPER METROID REDUX

1. In `$HOME/Applications/Distrobox/sm-redux`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`

2. Enter:
    - `make`
3. Wait for it to finish building
4. To play Super Metroid Redux, open `sm` in `$HOME/Applications/Distrobox/sm-redux/`

HOW TO CONFIGURE SUPER METROID REDUX

1. In `$HOME/Applications/Distrobox/sm-redux/`, right click `sm.ini`, click `Open with Kate` or a text editor of your choice
2. Customize settings

---

zelda3: A Link to the Past

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HOW TO COMPILE ZELDA3 ON THE STEAM DECK

WHAT IS ZELDA3?

`A reverse engineered clone of Zelda 3 - A Link to the Past.`

Source: https://github.com/snesrev/zelda3

INSTALLING PREREQUISITES

Note: In lieu of the following, you can simply download the zelda3 flatpak from the Discover store in Desktop Mode.

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following commands one line at a time:
    - `sudo apt install libsdl2-dev python3-pip make git python3-zstandard python3-yaml python3-pillow`

HOW TO SET UP ZELDA3

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone https://github.com/snesrev/zelda3`
3. A `zelda3` folder will be created, place your US Link to the Past ROM in `$HOME/Applications/zelda3`
    - SHA256 Hash: `66871d66be19ad2c34c927d6b14cd8eb6fc3181965b6e517cb361f7316009cfb`
    - To locate your SHA256 Hash, right click your ROM, click `Properties`, click `Checksums`, click `Calculate` to the left of `SHA1` and compare it to the above hash. If it is a match, you have a valid ROM
4. Rename the Link to the Past ROM to `zelda3.sfc`

HOW TO BUILD ZELDA3

1. In `$HOME/Applications/Distrobox/zelda3/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `make`
3. Wait for it to finish building
4. To play zelda3, open `zelda3` in `$HOME/Applications/Distrobox/zelda3/`

HOW TO CONFIGURE ZELDA3

1. In `$HOME/Applications/Distrobox/zelda3/`, right click `zelda3.ini`, click `Open with Kate` or a text editor of your choice
2. Customize settings

Recommended

- Set `ExtendedAspectRatio` to `16:10`
- Set `Fullscreen` to `1`

HOW TO CUSTOMIZE SPRITES

1. In `$HOME/Applications/Distrobox/zelda3/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone --recursive https://github.com/snesrev/sprites-gfx.git`
2. In `$HOME/Applications/Distrobox/zelda3/`, right click `zelda3.ini`, click `Open with Kate` or a text editor of your choice
3. Remove the `#` at the beginning of this line: `# LinkGraphics = sprites-gfx/snes/zelda3/link/sheets/megaman-x.2.zspr`
4. Replace `megaman-x.2.zspr` with the sprite of your choice

For a full list of sprites, visit: http://alttp.mymm1.com/sprites/

To find the file name to place in zelda3.ini

1. On http://alttp.mymm1.com/sprites/, right click the link above the sprite you would like to use, click `Copy Link`
2. In a new tab, paste the URL
    - If you press enter, it will download the sprite
3. The sprite file name will be at the end of the URL
4. Example (Crewmate): https://alttpr-assets.s3.us-east-2.amazonaws.com/crewmate.2.zspr
5. In `zelda3.ini`, replace `megaman-x.2.zspr` at the end: `LinkGraphics = sprites-gfx/snes/zelda3/link/sheets/megaman-x.2.zspr` with `crewmate2.zspr`
6. Example: `LinkGraphics = sprites-gfx/snes/zelda3/link/sheets/crewmate2.zspr`

HOW TO ENABLE MSU1 CD SOUNDTRACK FILES

Choose a soundtrack from: https://www.zeldix.net/t791-the-legend-of-zelda-a-link-to-the-past

Example

1. Download `Alternative New Soundtrack For Randomizers by JUD6MENT (updated: Sep 27, 2021)` from: https://www.zeldix.net/t791-the-legend-of-zelda-a-link-to-the-past to your `$HOME/Downloads` folder
2. Right click `Zelda Alternative Soundtrack by JUD6MENT (update Sep-21-2021).zip`, click `Extract archive here, autodetect subfolder`
3. Rename the newly extracted `Zelda Alternative Soundtrack by JUD6MENT (update Sep-21-2021)` folder to `msu`
    - `msu` is case sensitive
4. Move the `msu` folder to `$HOME/Applications/Distrobox/zelda3/`
5. In `$HOME/Applications/Distrobox/zelda3/`, right click `zelda3.ini`, click `Open with Kate` or a text editor of your choice
6. Edit the line: `EnableMSU = false` so it instead writes: `EnableMSU = true`
7. MSU1 CD Soundtrack Files are now enabled

Zelda64: Majora's Mask

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WHAT IS ZELDA64?

> **Info**
>
> Zelda 64: Recompiled is a project that uses N64: Recompiled to statically recompile Majora's Mask (and soon Ocarina of Time) into a native port with many new features and enhancements. This project uses RT64 as the rendering engine to provide some of these enhancements.

Source: https://github.com/Zelda64Recomp/Zelda64Recomp

HOW TO SET UP ZELDA64

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Zelda64` folder
2. Download the latest version of Zelda64, https://github.com/Zelda64Recomp/Zelda64Recomp/releases to `$HOME/Applications/Zelda64`
    - Download the `Zelda64Recompiled-v*.*.*-Linux-x64-AppImage.zip` file
3. Right click `Zelda64Recompiled-v*.*.*-Linux-x64-AppImage.zip`, click `Extract > Extract archive here`
    - If it creates a subfolder, move the contents directly to `$HOME/Applications/Zelda64`
4. Right click `Zelda64Recompiled-x86_64.AppImage`, click `Properties`, click `Permissions`, check `Is Executable`
5. Place a legally dumped **US** copy of `The Legend of Zelda: Majora's Mask` in `$HOME/Applications/Zelda64`
6. To open `Zelda64Recompiled-x86_64.AppImage`, double click it
7. Click `Select ROM` and select the ROM in `$HOME/Applications/Zelda64`

HOW TO CONFIGURE ZELDA64

1. Open `Zelda64Recompiled-x86_64.AppImage` at least once so it can generate the configuration files
2. Open the `$HOME/.config/Zelda64Recompiled` folder
    - `~/.config` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click `Show Hidden Files` to see these folders
3. Right click `general.json`, click `Open with Kate` or a text editor of your choice
4. Tweak settings
    - You may also tweak settings directly from the GUI

ZELDA64: SAVE LOCATION

Saves can be found in the `$HOME/.config/Zelda64Recompiled` folder. `~/.config` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click `Show Hidden Files` to see these folders.

4.1.5 General Maintenance

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How to Add Decompilations and Reverse Engineered PC Ports to Steam

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This section will use a simple script file to launch the various decompilations and reverse engineered ports on this page. You will need to create a script file per game.

- How to Create Script Files
- How to Utilize Script Files with ES-DE and Pegasus
- How to Utilize Script Files with Steam ROM Manager

HOW TO CREATE SCRIPT FILES

SCRIPT FILES: GROUP 1

- Cannonball OutRun Engine
- Render96ex
- sm64ex
- sm64ex-coop
- Sonic CD
- Sonic 1 and 2
- Super Mario Bros
- Super Mario Bros: The Lost Levels
- Super Mario World
- Super Metroid
- Super Metroid Redux
- zelda3: A Link to the Past

- Zelda64: Majora's Mask

1. In `Emulation/roms/desktop`, right click anywhere in the blank space, click `Create New` > `Text File` and give the text file a descriptive name (matching the game name typically) with a `.sh` file extension
   - For example: `Super Metroid.sh` or `The Legend of Zelda: A Link to the Past.sh`
2. Open the text file in a text editor of your choice
3. Enter the following (including the quotes):

```
#!/bin/bash
cd "/path/to/game"
"/path/to/gameexecutable"
```

4. Edit the path to where your game executable is located
   - For example:

```
#!/bin/bash
cd "$HOME/Applications/Distrobox/sm"
"$HOME/Applications/Distrobox/sm/sm"
```

5. Save and exit out of the file
6. Right click your newly created text file, click `Properties`, click `Permissions`, check `Is Executable`
7. You may now open the game by double clicking the newly created text file

Note:

- If your path is in the `home` directory, you may also replace `/home/$USER` (`home/deck` if you are on a Steam Deck) with `$HOME`
  - For example:

```
#!/bin/bash
cd "$HOME/Applications/Distrobox/sm"
"$HOME/Applications/Distrobox/sm/sm"
```

- For `Super Mario Bros` and `Super Mario Bros: The Lost Levels`, you will also need to include the path to the respective game file in addition to the `smw` executable. For more information, see How to Play Super Mario Bros and How to Play Super Mario Bros: The Lost Levels.

SCRIPT FILES: GROUP 2

- Super Mario 64 Plus

1. Download attached `.sh` file
   - SuperMario64Plus.sh
   - **Note:** If you are using different folder locations, make sure to edit the above file and edit the paths
2. Place in `Emulation/roms/desktop`
3. Right click your newly created text file, click `Properties`, click `Permissions`, check `Is Executable`
4. You may now open the game by double clicking the newly created text file

SCRIPT FILES: GROUP 3

- Perfect Dark

1. Download attached `.sh` file
   - PerfectDark.sh
   - **Note:** If you are using different folder locations, make sure to edit the above file and edit the paths
2. Place in `Emulation/roms/desktop`
3. Right click your newly created text file, click `Properties`, click `Permissions`, check `Is Executable`
4. Use `PerfectDark.sh` to open Perfect Dark

SCRIPT FILES: GROUP 4

- 2Ship2Harkinian: Majora's Mask
- Ship of Harkinian: Ocarina of Time

1. In a folder of your choice, create a text file and give it a descriptive name (matching the game name typically) with a `.sh` file extension
   - For example: `Ship of Harkinian.sh`
2. Open the text file in a text editor of your choice
3. Enter the following:

```
#!/bin/bash

cd "/path/to/folderofgameexecutable"

"/path/to/gameexecutable"
```

4. Edit the paths to where your game executable is located
   - For example:

```
#!/bin/bash

cd "$HOME/Applications/ShipofHarkinian"

"$HOME/Applications/ShipofHarkinian/soh.appimage"
```

5. Save and exit out of the file
6. Right click your newly created text file, click `Properties`, click `Permissions`, check `Is Executable`

Note:

- If your path is in the `home` directory, you may also replace `/home/$USER` (`home/deck` if you are on a Steam Deck) with `$HOME`
  - For example:

```
#!/bin/bash

cd "$HOME/Applications/ShipofHarkinian"

"$HOME/Applications/ShipofHarkinian/soh.appimage"
```

SCRIPT FILES: GROUP 5

- OpenGOAL: Jak and Dexter: The Precursor Legacy
- OpenGOAL: Jak II

1. In `Emulation/roms/desktop`, right click anywhere in the blank space, click `Create New` > `Text File`
2. Name it matching the respective Jak and Dexter game
3. At the top of the text file, write `#!/bin/bash`
4. In `$HOME/Applications/OpenGOAL`, double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it
5. On the left side, select the matching Jak and Dexter game you created the script file for in Step 2
6. On the bottom right, click the `Gear` icon, and click `Copy Game Executable Command`
7. In the text file you created in Step 1, right click anywhere below `#!/bin/bash` and click `Paste`
8. Save the text file and exit out
9. Right click your newly created text file, click `Properties`, click `Permissions`, check `Is Executable`
10. You may now open the game by double clicking the newly created text file

Note: You will need to update this shortcut whenever you update OpenGOAL.

HOW TO UTILIZE SCRIPT FILES WITH ES-DE AND PEGASUS

1. Place your script files in `Emulation/roms/desktop`

2. In Game Mode, open ES-DE or Pegasus and play your newly created script files

HOW TO UTILIZE SCRIPT FILES WITH STEAM ROM MANAGER

1. Place your script files in `Emulation/roms/desktop`
2. In `$HOME/.config/steam-rom-manager/userData/`, open `userConfigurations.json` in a text editor of your choice
3. Scroll to the very bottom of the text file, you will see a `}` and a `]`, add a comma to `}`
    - Steam ROM Manager Parser
4. Paste the below block of text between the `},` and the `]`

```
	{
	   "parserType": "Glob",
	   "configTitle": "Decompilations",
	   "steamDirectory": "${steamdirglobal}",
	   "steamCategory": "${Decompilations}",
	   "romDirectory": "${romsdirglobal}/desktop",
	   "executableArgs": "",
	   "executableModifier": "\"${exePath}\"",
	   "startInDirectory": "",
	   "titleModifier": "${fuzzyTitle}",
	   "fetchControllerTemplatesButton": null,
	   "removeControllersButton": null,
	   "imageProviders": [
	         "SteamGridDB"
	   ],
	   "onlineImageQueries": "${${fuzzyTitle}}",
	   "imagePool": "${fuzzyTitle}",
	   "userAccounts": {
	         "specifiedAccounts": null
	   },
	   "executable": {
	         "path": "",
	         "shortcutPassthrough": true,
	         "appendArgsToExecutable": true
	   },
	   "parserInputs": {
	         "glob": "**/${title}@(.sh|.SH|.desktop|.DESKTOP)"
	   },
	   "titleFromVariable": {
	         "limitToGroups": "",
	         "caseInsensitiveVariables": false,
	         "skipFileIfVariableWasNotFound": false,
	         "tryToMatchTitle": false
	   },
	   "fuzzyMatch": {
	         "replaceDiacritics": true,
	         "removeCharacters": true,
	         "removeBrackets": true
	   },
	   "controllers": {
	         "ps4": null,
	         "ps5": null,
	         "xbox360": null,
	         "xboxone": null,
	         "switch_joycon_left": null,
	         "switch_joycon_right": null,
	         "switch_pro": null,
	         "neptune": null
	   },
	   "imageProviderAPIs": {
	         "SteamGridDB": {
	            "nsfw": false,
	            "humor": false,
	            "styles": [],
	            "stylesHero": [],
	            "stylesLogo": [],
	            "stylesIcon": [],
	            "imageMotionTypes": [
	               "static"
	            ]
	         }
	   },
	   "defaultImage": {
	         "tall": null,
	         "long": null,
	         "hero": null,
	         "logo": null,
	         "icon": null
	   },
	   "localImages": {
	         "tall": null,
	         "long": null,
	         "hero": null,
	         "logo": null,
	         "icon": null
	   },
	   "parserId": "168816977280299277",
```

```
        "version": 15
    }
```

5. Open Steam ROM Manager, toggle the `Decompilations` parser and generate an app list to add your games to Steam

---

How to Update Decompilations and Reverse Engineered PC Ports

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UPDATING: GROUP 1

- Render96ex
- Super Mario 64 Plus
- sm64ex
- sm64ex-coop
- Sonic 1 and 2
- Sonic CD
- Super Mario Bros
- Super Mario Bros: The Lost Levels
- Super Mario World
- Super Metroid

- Super Metroid Redux
- zelda3: A Link to the Past

1. In the respective folder, open a terminal and enter:
    - `git pull origin main`
    - If this does not work, try:

    ```
    * `git pull origin master`
    ```

2. Resolve any conflicts
    - If it states an ini file is in conflict, rename the pre-existing ini file and add a `.bak` at the end of the file name

    ```
    * For example: `smw.ini.bak`
    ```

3. Re-build the game
    - You may need to do `make clean` first using a terminal in the root of the project folder
    - You may reference the various sections to re-build the game. The steps will be identical

UPDATING: GROUP 2

- Perfect Dark

1. Download the `i686-linux` version from https://github.com/fgsfdsfgs/perfect_dark#download to your `Downloads` folder
2. Right click `pd-i686-linux.zip`, click `Extract archive here, autodetect subfolder`
3. Move the newly extracted `pd-i686-linux` folder to `$HOME/Applications`
4. Overwrite the current `pd` file
5. Right click `pd`
6. Click `Properties`
7. Click `Permissions`
8. Check `Is Executable`

UPDATING: GROUP 3

- 2Ship2Harkinian: Majora's Mask
- Ship of Harkinian: Ocarina of Time

2Ship2Harkinian

1. Download the latest `Linux` version of `2Ship2Harkinian`: https://github.com/HarbourMasters/2ship2harkinian/releases to `$HOME/Applications/2Ship2Harkinian`
    - Download the `2Ship-VERSION-NAME-Linux.zip` file
2. Right click the downloaded zip file and click `Extract > Extract archive here`
3. Overwrite the current `2ship.appimage` file
4. Right click `2ship.appimage`, click `Properties`, click `Permissions`, check `Is Executable`
5. When a new OTR is required, delete the current OTR files in `$HOME/Applications/2Ship2Harkinian` and re-run the `2ship.appimage` file by double clicking it to generate new OTR files
    - See https://www.shipofharkinian.com/faq#what-do-the-release-numbers-mean for details

Ship of Harkinian: Ocarina of Time

1. Download the latest `Linux-Performance` version of `Ship of Harkinian`: https://github.com/HarbourMasters/Shipwright/releases to `$HOME/Applications/ShipofHarkinian`
    - Download the `SoH-VERSION-NAME-Linux-Performance.zip` file
2. Right click the downloaded zip file and click `Extract > Extract archive here`

3. Overwrite the current `soh.AppImage` file
4. Right click `soh.AppImage`, click `Properties`, click `Permissions`, check `Is Executable`
5. When a new OTR is required, delete the current OTR files in `$HOME/Applications/ShipofHarkinian` and re-run the `soh.AppImage` file by double clicking it to generate new OTR files
    - See https://www.shipofharkinian.com/faq#what-do-the-release-numbers-mean for details

UPDATING: GROUP 4

- OpenGOAL: Jak and Dexter: The Precursor Legacy
- OpenGOAL: Jak II

1. In `$HOME/Applications/OpenGOAL`, double click `open-goal-launcher_#.#.#_amd64.AppImage` to open it
2. On the left, select the `Gear` icon, click `Versions` at the top, click the `Download` icon on the top-most row (usually will also have the most recent date)
3. Click the circle icon on the left to select the version and click `Save` in the top right
4. Click the game you would like to update on the left
5. You will receive a message requesting you to update the game, click `Update Game` and wait a few moments
6. Your game will now be updated to the latest version

Note: Do not forget to update the shortcut as well whenever you update OpenGOAL.

UPDATING: GROUP 5

- Zelda64: Majora's Mask

1. Download the latest version of Zelda64, https://github.com/Zelda64Recomp/Zelda64Recomp/releases to `$HOME/Applications/Zelda64`
    - Download the `Zelda64Recompiled-v*.*.*-Linux-x64-AppImage.zip` file
2. Right click `Zelda64Recompiled-v*.*.*-Linux-x64-AppImage.zip`, click `Extract > Extract archive here`
    - **Overwrite** the current AppImage
    - If it creates a subfolder, move the contents directly to `$HOME/Applications/Zelda64`
    - If you do not overwrite the outdated AppImage, simply delete it instead
3. Right click `Zelda64Recompiled-x86_64.AppImage`, click `Properties`, click `Permissions`, check `Is Executable`

4.1.6 Emulators

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BlueMaxima's Flashpoint

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Link: https://gist.github.com/parkerlreed/4bd1f5fa38f7ffa72f9ceacb7d7f636d

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Hypseus Singe

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Link: https://gitlab.com/es-de/emulationstation-de/-/blob/master/USERGUIDE.md#hypseus-singe-daphne

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4.1.7 Emulation Related Games

AM2R

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WHAT IS AM2R?

`It is an unofficial remake of the 1991 Game Boy game Metroid II: Return of Samus in the style of Metroid: Zero Mission (2004).`

Source: https://en.wikipedia.org/wiki/AM2R

HOW TO PLAY AM2R

Note: You will need a copy of `AM2R_1_1.zip`, this section will not cover how to obtain that file.

1. In Desktop Mode, open Discover
2. Search for AM2R
    - Flathub page: https://flathub.org/apps/io.github.am2r_community_developers.AM2RLauncher

    ```
    * Discover (on the Steam Deck) uses Flathub as its source for applications
    ```

3. Download AM2R
4. Open AM2R and click the big `Download` button on the main screen
5. Select `Select AM2R_1_1.zip`, select your file, and play the game

HOW TO ADD AM2R TO STEAM

1. Download attached `.sh` file
    - AM2R.sh
2. Place in `$HOME/Applications`
3. Right click `AM2R.sh`
4. Click `Properties`
5. Click `Permissions`
6. Check `Is Executable`
7. Use `AM2R.sh` to open AM2R
8. In Desktop Mode, right click `AM2R.sh`, click `Add to Steam`
    - Alternatively, place `AM2R.sh` in `Emulation/roms/desktop` and use EmulationStation-DE or the Steam ROM Manager parser here: How to Utilize Script Files with Steam ROM Manager

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Link's Awakening DX HD

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WHAT IS LINK'S AWAKENING DX HD?

`Dive into the enchanting world of The Legend of Zelda: Link's Awakening DX as you've never experienced it before, with this meticulously crafted PC version that breathes new life into this classic adventure. Immerse yourself in the nostalgia of Koholint Island with enhanced graphics and widescreen support, bringing the charming landscapes and characters to vivid detail on your modern PC display.`

Source: https://gbatemp.net/threads/fanmade-pc-port-of-the-legend-of-zelda-links-awakening-remakes-the-game-in-hd-with-widescreen-support.644566/

HOW TO SET UP LINK'S AWAKENING DX HD

Note: You will need a copy of `Link's Awakening DX HD`, this section will not cover how to locate it. Shortly after Link's Awakening DX HD released, it received a takedown notice by Nintendo and its official itch.io page with the game download was shut down.

Link's Awakening DX HD

1. Open `Konsole`
2. Enter:
    - `mkdir -p $HOME/Games/Links_Awakening_DX_HD/pfx`
    - This command will create a couple of **empty** folders to make managing and installing Link's Awakening DX HD easier
3. Place your game files directly in `$HOME/Games/Links_Awakening_DX_HD`
    - If you are on a Steam Deck, this path may be `$HOME/Games/Links_Awakening_DX_HD`

Lutris

1. In Desktop Mode, open Discover
2. Search for Lutris
    - Flathub page: https://flathub.org/apps/net.lutris.Lutris

    ```
    * Discover (on the Steam Deck) uses Flathub as its source for applications
    ```

3. Download Lutris
4. Open Lutris and click the `Wine` button on the left, click the `Manage versions` button
5. Download `wine-ge-8-25`
6. Close out of the `Manage versions` menu and click the `+` button in the top left
7. Click `Add locally installed game`
8. `Game Info` tab:
    - Name: `Link's Awakening DX HD`
    - Sort name: Leave blank
    - Runner: `Wine (runs Windows games)`
    - Release year: Leave blank
9. Game options:
    - Executable: Click the `Browse` button and navigate to the `Link's Awakening DX HD.exe` file in `$HOME/Games/Links_Awakening_DX_HD`

    ```
    * If you are on a Steam Deck, this path may be `$HOME/Games/Links_Awakening_DX_HD/Link's Awakening DX HD.exe`
    ```

    - Arguments: Leave blank
    - Working directory: Leave blank
    - Wine prefix: Click the `Browse` button and select the `pfx` folder in `$HOME/Games/Links_Awakening_DX_HD`

    ```
    * If you are on a Steam Deck, this path may be `$HOME/Games/Links_Awakening_DX_HD/pfx`
    ```

    - Prefix architecture: `Auto (default)`
10. Runner options:
    - Wine version: `wine-ge-8-25-x86-64`
    - You do not need to adjust any additional settings on this tab
11. Click `Save` in the top right
12. Single click `Link's Awakening DX HD` in Lutris and click the `Wine` icon at the bottom of the screen, click `Winetricks`
13. Click `Select the default wineprefix` and click `OK`
14. Select `Install a Windows DLL or component` and click `OK`

15. Select `dotnetdesktop6` under the `Package` column and click `OK`, wait for it to finish installing
16. Close out of winetricks
17. To play Link's Awakening DX HD, open Lutris, click the Link's Awakening DX HD square and click `Play` in the bottom middle of the screen

HOW TO ADD LINK'S AWAKENING DX HD TO STEAM

1. Open Lutris
2. Right click Link's Awakening DX HD, click `Create desktop shortcut`
3. A shortcut of Link's Awakening DX HD will be added to your desktop
4. On your desktop, right click `Link's Awakening DX HD` and click `Add to Steam`
    - Alternatively, move this desktop file to `Emulation/roms/desktop` and use EmulationStation-DE or the Steam ROM Manager parser here: How to Utilize Script Files with Steam ROM Manager

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PokeMMO

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Note: You will need a legal copy of the following games:

- Pokemon Black or Pokemon White
- Optional:
    - Pokemon Fire Red
    - Pokemon Platinum
    - Pokemon Emerald
    - Pokemon HeartGold or Pokemon SoulSilver

HOW TO INSTALL POKEMMO

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In the `$HOME/Applications` folder, create a `PokeMMO` folder
2. Install Java, for instructions see How To Install Java on the Steam Deck
3. Download PokeMMO: https://pokemmo.com/downloads/linux/ to `$HOME/Games/PokeMMO`
    - Under `Other Distributions`, click `Download the Client`
4. Extract `PokeMMO-Client.zip` to `$HOME/Games/PokeMMO`
    - If it creates a subfolder, move the contents directly to `$HOME/Games/PokeMMO`
5. Right click `PokeMMO.sh`, click `Properties`, click `Permissions`, check `Is Executable`
6. Right click `PokeMMO.sh`, click `Open with Kate` or a text editor of your choice
7. At the top of the file, below `#!/bin/sh`, write the following two lines:

```
export JAVA_HOME=$HOME/Applications/jdk
export PATH=$JAVA_HOME/bin:$PATH
```

8. Save and exit out of the file
9. To open PokeMMO, double click `PokeMMO.sh`

HOW TO ADD POKEMMO TO STEAM

1. In Desktop Mode, right click `PokeMMO.sh` in `$HOME/Games/PokeMMO`, click `Add to Steam`
    - Alternatively, use the Steam ROM Manager parser here: How to Utilize Script Files with Steam ROM Manager to add it to Steam

---

Sonic 3 A.I.R.

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PREFACE

To play Sonic 3 A.I.R., you will need an original and legal copy of Sonic 3 & Knuckles from the Steam version of Sonic 3 & Knuckles. This section will not cover how to obtain that file. For more information, see https://docs.google.com/document/d/1oSud8dJHvdfrYbkGCflIAOp3JuTks7z4K5SwtVkXkx0/edit#heading=h.ux87lw254eyd.

Sonic 3 A.I.R. has a comprehensive manual covering many of the features and settings in the game. See https://sonic3air.org/Manual.pdf.

WHAT IS SONIC 3 A.I.R.

`A fan-made widescreen remaster of Sonic 3 & Knuckles.`

Source: https://sonic3air.org/

SETTING UP SONIC 3 A.I.R.

1. In Desktop Mode, open Discover
2. Search for Sonic 3: Angel Island Revisited
    - Flathub page: https://flathub.org/apps/org.sonic3air.Sonic3AIR

    ```
    * Discover (on the Steam Deck) uses Flathub as its source for applications
    ```

3. Download Sonic 3: Angel Island Revisited
4. Open Sonic 3: Angel Island Revisited
5. Select your `Sonic 3 & Knuckles` game file
    - The Steam version is typically named: `Sonic_Knuckles_wSonic3.bin`
    - For more information, read the Linux section on the manual: https://sonic3air.org/Manual.pdf

`Sonic 3: Angel Island Revisited` Folder and File Locations

`$HOME/.var/app/org.sonic3air.Sonic3AIR`

```
org.sonic3air.Sonic3AIR/
├── cache
├── config
└── data
    └── Sonic3AIR
        ├── logfile.txt
        ├── mods
        ├── settings_input.json
        ├── settings.json
        └── Sonic_Knuckles_wSonic3.bin
```

HOW TO CONFIGURE SONIC 3 A.I.R.

1. Open the `$HOME/.var/app/org.sonic3air.Sonic3AIR/data/Sonic3AIR` folder
    - `~/.var` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click `Show Hidden Files` to see these folders
2. Right click `settings.json` or `settings_input.json`, click `Open with Kate` or a text editor of your choice
    - `settings.json` is specifically for game settings
    - `settings_input.json` is specifically for game controls
3. Customize settings

HOW TO MOD SONIC 3 A.I.R.

Mod Resources

This list is not comprehensive

- https://sonic3air.boards.net/board/6/mod-releases
- https://gamebanana.com/mods/games/6878

1. Open the `$HOME/.var/app/org.sonic3air.Sonic3AIR/data/Sonic3AIR` folder
    - `~/.var` is a hidden folder by default. In Dolphin (file manager), click the hamburger menu in the top right, click `Show Hidden Files` to see these folders
2. Download and extract mod(s) to the `mod` folder
    - A mod folder should typically have a `Data` folder and a `mod.ini` file
3. To activate/toggle the mod(s), open Sonic 3 A.I.R.. On the main menu, click the `Mods` button

HOW TO ADD SONIC 3 A.I.R. TO STEAM

1. Download attached `.sh` file
    - Sonic3AIR.sh
2. Place in `$HOME/Applications`
3. Right click `Sonic3AIR.sh`
4. Click `Properties`
5. Click `Permissions`
6. Check `Is Executable`
7. Use `Sonic3AIR.sh` to open Sonic 3 A.I.R.
8. In Desktop Mode, right click `Sonic3AIR.sh`, click `Add to Steam`
    - Alternatively, use the Steam ROM Manager parser here: How to Utilize Script Files with Steam ROM Manager to add it to Steam

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4.1.8 Emulation Related Tools

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Skyscraper

WHAT IS THIS?

> **Quote**
>
> A powerful and versatile yet easy to use game scraper written in C++ for use with multiple frontends running on a Linux system (macOS and Windows too, but not officially supported). It scrapes and caches various game resources from various scraping sources, including media such as screenshot, cover and video. It then gives you the option to generate a game list and artwork for the chosen frontend by combining all of the cached resources.

Source: https://github.com/Gemba/skyscraper

INSTALLING PREREQUISITES

1. To do the steps in this section, you will need to have created an Ubuntu Distrobox. If you have not created one already, see How to Set up Distrobox for instructions
2. Enter the Distrobox by opening Konsole and entering:
    - `distrobox enter ubuntu`
3. Enter the following command:
    - `sudo apt install qtbase5-dev qtchooser qt5-qmake qtbase5-dev-tools p7zip-full build-essential`

HOW TO SET UP SKYSCRAPER

Note: The following folder locations are recommendations. You may choose a different folder location.

1. In `$HOME/Applications`, create a `Distrobox` folder
2. In the `Distrobox` folder, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `git clone https://github.com/Gemba/skyscraper.git`
3. A `skyscraper` folder will be created

HOW TO BUILD SKYSCRAPER

1. In `$HOME/Applications/Distrobox/skyscraper/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `./update_skyscraper.sh`
3. Wait for it to finish building
4. To open Skyscraper, open `Skyscraper` in `$HOME/Applications/Distrobox/skyscraper/` using a terminal
    - For example, right click anywhere in `$HOME/Applications/Distrobox/skyscraper/`, click `Open Terminal Here`, and type `./Skyscraper`
    - You **do** not need to enter the Distrobox again to open Skyscraper

HOW TO CONFIGURE SKYSCRAPER

1. In `$HOME/.skyscraper`, right click `config.ini`, click `Open with Kate` or a text editor of your choice
2. Customize settings

HOW TO UPDATE SKYSCRAPER

1. In `$HOME/Applications/Distrobox/skyscraper/`, right click anywhere in the folder, click `Open Terminal Here`, enter:
    - `distrobox enter ubuntu`
2. Enter:
    - `./update_skyscraper.sh`
3. Wait for it to finish updating

SKYSCRAPER USAGE

For more detailed information on how to use Skyscraper, see the Scraping section on the Pegasus page.

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5.1 Compression Tools

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5.3 Compiling Software

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5.4 Distrobox

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6. Emulators

6.1 Emulators